PROGNOSTIC FACTORS FOR STREPTOCOCCAL TOXIC SHOCK SYNDROME: SYSTEMATIC REVIEW AND META-ANALYSIS

Supplementary Material

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Search strategies

1) Database: OVID Medline Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily and Ovid MEDLINE(R) 1946 to Present

Search Strategy:
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1 toxic shock syndrome.mp. or exp Shock, Septic/ or STSS.mp. or streptococcal toxic shock syndrome.mp. or necrotizing fasciitis.mp. or exp Fasciitis, Necrotizing/ or septic shock.mp.
2 (group a streptococc* or group A streptococc* or pyogenes or streptococcus pyogenes).mp. or exp Streptococcus pyogenes/
3 exp Cohort Studies/
4 cohort$.tw.
5 controlled clinical trial.pt.
6 epidemiologic methods/
7 limit 6 to yr=1966-1989
8 exp case-control studies/
9 (case$ and control$).tw.
10 (case$ and series).tw.
11 or/3-5,7-10
12 randomized controlled trial.pt.
13 (random$ or placebo$ or single blind$ or double blind$ or triple blind$).ti,ab.
14 (retraction of publication or retracted publication).pt.
15 or/12-14
16 (animals not humans).sh.
17 ((comment or editorial or meta-analysis or practice-guideline or review or letter) not randomized controlled trial).pt.
18 (random sampl$ or random digit$ or random effect$ or random survey or random regression).ti,ab. not randomized controlled trial.pt.
19 15 not (16 or 17 or 18)
20 animals/ not humans/
21 (1 or 2) and (11 or 19)
22 21 not 20

2) Database: Embase <1974 to 2020 February 03>

Search Strategy:
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1 toxic shock syndrome.mp. or exp septic shock/ or STSS.mp. or streptococcal toxic shock syndrome.mp. or necrotizing fasciitis.mp. or exp necrotizing fasciitis/ or septic shock.mp. or exp toxic shock syndrome/
2 (group a streptococc* or group A streptococc* or pyogenes or streptococcus pyogenes).mp. or exp Streptococcus pyogenes/ or exp Streptococcus group A/
3 exp cohort analysis/
4 exp longitudinal study/
5 exp prospective study/

6 exp follow up/
7 cohort$.tw.
8 exp case control study/ or (case$ and control$).tw.
9 exp case study/ or (case$ and series).tw.
10 or/3-9
11 (random$ or placebo$ or single blind$ or double blind$ or triple blind$).ti,ab.
12 RETRACTED ARTICLE/
13 or/11-12
14 (animal$ not human$).sh,hw.
15 (book or conference paper or editorial or letter or review).pt. not exp randomized controlled trial/
16 (random sampl$ or random digit$ or random effect$ or random survey or random regression).ti,ab. not exp randomized controlled trial/
17 13 not (14 or 15 or 16)
18 exp animal/
19 exp human/
20 18 not 19
21 (1 or 2) and (10 or 17)
22 21 not 20

GRADE assessment guidance

Based on GRADE guidance for prognostic studies, we start with high certainty evidence for each meta-analysis.

Risk of bias
For each meta-analysis, we downgraded the certainty of the evidence once for risk of bias when studies with moderate or high risk of bias contributed >50% weight to the pooled effect estimates or when studies providing unadjusted estimates contributed >50% weight to the pooled effect estimate (i.e. serious risk of bias). We downgraded the certainty of the evidence twice for risk of bias when studies with moderate or high risk of bias contributed >80% weight to the pooled effect estimates or when studies providing unadjusted estimates contributed >80% weight to the pooled effect estimate (i.e. very serious risk of bias).

Inconsistency
We used visual inspection of the forest plots and the I^2 statistic to assess inconsistency. For visual inspection of the forest plots, we considered the variability in point estimates and confidence interval overlap in relation to the null effect. Further, we downgraded the certainty of the evidence once when there was substantial (I^2 50-90%) heterogeneity and twice when there was considerable (I^2 75-100%) heterogeneity.

Imprecision
We downgraded the certainty of the evidence once for imprecision if:
1) The effect on the patient, or clinical action, would differ depending on whether the upper or the lower boundary of the confidence interval represented the truth, **OR**
2) The 95% CI of the pooled absolute estimate crossed the null effect by less than 50 people per 1000 on one or both sides, **OR**
3) In cases where we had large effects that did not cross the null, we assessed the optimal information size. If the ratio of the upper to the lower limit of the confidence interval of the relative estimate was >3, the optimal information size would never be met; thus, we rated down once for imprecision, **OR**
4) There were fewer than 10 outcome events for each prognostic variable (for dichotomous outcomes, **OR**
5) There were fewer than 100 cases reaching endpoint (for continuous outcomes).
We downgraded the certainty of the evidence twice for imprecision if:
1) The 95% CI of the pooled absolute estimate crossed the null effect by more than 50 people per 1000 on both sides.

Indirectness
We rated down once if the study sample, the prognostic factor, and/or the outcome in the primary studies did not accurately reflect the review question.

Publication bias
We rated down once if:
1) Small studies reported higher rates compared to large studies, suggesting the selective publication of "positive" studies, **OR**

2) The value of the risk/protective factor in predicting the outcome has NOT been repetitively investigated (e.g. only exploratory studies with no external validation, replication or confirmation exist).

References

1. Guyatt GH, Oxman AD, Kunz R, et al. GRADE Guidelines 6. Rating the Quality of Evidence—Imprecision. J Clin Epidemiol 2011; 64(12): 1283–93. https://doi.org/10.1016/j.jclinepi.2011.01.012
2. Huguet A, Hayden JA, Stinson J, et al. Judging the quality of evidence in reviews of prognostic factor research: adapting the GRADE framework. Syst Rev. 2013;2:71. Published 2013 Sep 5. doi:10.1186/2046-4053-2-71

Table of excluded full texts (n=264)

Author, Year	Title	Reason for Exclusion
Hamilton, 2013	Pregnancy-related group a streptococcal infections: Temporal relationships between bacterial acquisition, infection onset, clinical findings, and outcome	Wrong study design
Ichiyama, 1997	Transmission of Streptococcus pyogenes causing toxic shock-like syndrome among family members and confirmation by DNA macrorestriction analysis	Wrong study design
Idubor, 2019	Invasive group a streptococcus infections among residents of multiple nursing Homes-Denver, Colorado, 2017-2018	Wrong study design
Ikebe, 2015	Increased prevalence of group A streptococcus isolates in streptococcal toxic shock syndrome cases in Japan from 2010 to 2012	Wrong study design
Klinker, 1997	Toxic shock syndrome in AIDS	Wrong study design
Langmuir, 1982	Toxic-shock syndrome--an epidemiologist's view	Wrong study design
McIvor, 1982	Treatment of recurrent toxic shock syndrome with oral contraceptive agents	Wrong study design
Pathi, 2013	Prompt recognition and multidisciplinary approach in Group A streptococcal sepsis	Wrong study design
Shah, 2015	Role of intravenous immune globulin in streptococcal toxic shock syndrome and Clostridium difficile infection	Wrong study design
Stallones, 1982	A review of the epidemiologic studies of toxic shock syndrome	Wrong study design
Stevens, 2000	Molecular epidemiology of nga and NAD glycohydrolase/ADP-ribosyltransferase activity among Streptococcus pyogenes causing streptococcal toxic shock syndrome	Wrong study design
Turner, 2015	Emergence of a New Highly Successful Acapsular Group A Streptococcus Clade of Genotype emm89 in the United Kingdom	Wrong study design
Udagawa, 1999	Serious group A streptococcal infection around delivery	Wrong study design
Valiquette, 2006	A survey of physician's attitudes regarding management of severe group A streptococcal infections	Wrong study design
Valiquette, 2009	Assessing the impact of intravenous immunoglobulin in the management of streptococcal toxic shock syndrome: a noble but difficult quest	Wrong study design
Wozniak, 2012	M-protein gene-type distribution and hyaluronic acid capsule in group A Streptococcus clinical isolates in Chile: Association of emm gene markers with csrR alleles	Wrong study design
Anonymous, 1984	Diagnostic bias and toxic shock syndrome	Wrong study design
Blonde, 2017	A prospective multicentric study of severe cutaneous infections in pediatric intensive care: The SCIPIC cohort	Wrong study design
Busowski, 2010	Puerperal group A streptococci (GAS) infection: Reemergence of a dreaded disease - A case series	Wrong study design
Cimolai, 2002	Nonhemolytic Streptococcus pyogenes causing invasive infection	Wrong study design
Clark, 2010	Puerperal group a streptococcal sepsis and proinflammatory immune response	Wrong study design

Dahdi, 2018	Necrotizing soft tissue infection: Microbiological distribution from district region	Wrong study design
Das, 2012	Necrotizing fasciitis in New Zealand - Risk factors, microbiological findings and outcomes in a large case series	Wrong study design
De Zoysa, 2013	Invasive group A streptococcal disease in the UK, 2008-2012 and molecular characterisation of isolates during enhanced surveillance	Wrong study design
Donawa, 1984	Toxic shock syndrome: chronology of state and federal epidemiologic studies and regulatory decision-making	Wrong study design
Figueira, 2013	Peri-ocular necrotising fasciitis: A multicentre retrospective australian series	Wrong study design
Gaensbauer, 2016	Importance of toxic shock syndrome in pediatric septic shock clinical decision-making	Wrong study design
Goldberg, 2015	Group A beta streptococcal infections in children after oral or dental trauma: A case series of 5 patients	Wrong study design
McViety, 2014	Don't forget IGAS: Lessons learnt from review of 2 peak seasons in the north west and North Wales, UK	Wrong study design
Zangara, 2019	Epidemiology, outcomes from treatment, and the spectrum of soft tissue infections over time in hospitalized patients: A populationbased description of inpatients in the state of california	Wrong study design
Arias-Constanti, 2018	Invasive disease by Streptococcus pyogenes: patients hospitalized for 6 years	Wrong population
Hankins, 2008	Factors that affect the clinical course of group A beta-haemolytic streptococcal infections of the hand and upper extremity: a retrospective study	Wrong population
Henrichsen, 1997	Invasive infections caused by Streptococcus pyogenes in Denmark 1990- 1994	Wrong population
Hoge, 1993	The changing epidemiology of invasive group A streptococcal infections and the emergence of streptococcal toxic shock-like syndrome. A retrospective population-based study	Wrong population
Jauregui, 2015	Life- and limb-threatening infections following the use of an external fixator	Wrong population
Kadri, 2017	Impact of Intravenous Immunoglobulin on Survival in Necrotizing Fasciitis With Vasopressor-Dependent Shock: A Propensity Score-Matched Analysis From 130 US Hospitals	Wrong population
Leggiadro, 1993	Group A streptococcal bacteremia in a mid-south children's hospital	Wrong population
Madsen, 2019	Patient's characteristics and outcomes in necrotising soft-tissue infections: results from a Scandinavian, multicentre, prospective cohort study	Wrong population
Mitchell, 2011	A strep in the wrong direction-invasive group a streptococcal disease	Wrong population
Moses, 1995	Group A streptococcus bacteremia at the Hadassah Medical Center in Jerusalem	Wrong population
Mosites, 2017	Use of single-dose azithromycin to control a community outbreak of EMM26.3 group a streptococcus invasive disease-Alaska, 2017	Wrong population
Mosites, 2019	Risk for invasive streptococcal infections among adults experiencing homelessness, anchorage, Alaska, USA, 2002-2015	Wrong population

Mulla, 2007	Clinical and epidemiologic features of invasive group A streptococcal infections in children	Wrong population
Mulla, 2003	Invasive group A streptococcal infections in Florida	Wrong population
Navarro, 1993	A comparison of Streptococcus pyogenes (group A streptococcal) bacteremia at an urban and a suburban hospital. The importance of intravenous drug use	Wrong population
Norton, 2004	Invasive group A streptococcal disease in North Queensland (1996 - 2001)	Wrong population
Nuwayhid, 2007	Blunt Trauma as a Risk Factor for Group A Streptococcal Necrotizing Fasciitis	Wrong population
Oliver, 2019	Invasive group A Streptococcus disease in Australian children: 2016 to 2018 - a descriptive cohort study	Wrong population
Oliver, 2019	Recent trends in invasive group A Streptococcus disease in Victoria	Wrong population
Peterson, 1996	Risk factors for invasive group A streptococcal infections in children with varicella: A case-control study	Wrong population
Rainbow, 2008	Invasive group A streptococcal disease in nursing homes, Minnesota, 1995-2006	Wrong population
Rathore, 1992	Suppurative group A beta-hemolytic streptococcal infections in children	Wrong population
Reingold, 1984	Epidemiology of toxic-shock syndrome, United States, 1960-1984	Wrong population
Reingold, 1989	Risk factors for menstrual toxic shock syndrome: results of a multistate case-control study	Wrong population
Remy, 2019	Septic shock and toxic shock syndrome-two infectious shocks with different immune response	Wrong population
Rudolph, 2016	Epidemiology of invasive group a streptococcal disease in Alaska, 2001 to 2013	Wrong population
Saldana, 2010	Periorbital necrotizing fasciitis: Outcomes using a CT-guided surgical debridement approach	Wrong population
Sarangi, 1995	A nursing home outbreak of group A streptococcal infection: case control study of environmental contamination	Wrong population
Scaber, 2011	Group a streptococcal infections during the seasonal influenza outbreak 2010/11 in South East England	Wrong population
Scheuer, 2018	Rising high rate of invasive group a streptococcus infections among persons experiencing homelessness in San Francisco, 2010-2017	Wrong population
Schlech, 1982	Risk factors for development of toxic shock syndrome. Association with a tampon brand	Wrong population
Schwartz, 1989	Nonmenstrual toxic shock syndrome associated with barrier contraceptives: report of a case-control study	Wrong population
Shands, 1982	Toxic shock syndrome: Case-control studies at the centers for disease control	Wrong population
Sharma, 2019	Real-time whole genome sequencing to control a Streptococcus pyogenes outbreak at a national orthopaedic hospital	Wrong population
Sierra, 2006	Group A streptococcal infections in injection drug users in Barcelona, Spain: Epidemiologic, clinical, and microbiologic analysis of 3 clusters of cases from 2000 to 2003	Wrong population

Simonart, 2004	The importance of serum creatine phosphokinase level in the early diagnosis and microbiological evaluation of necrotizing fasciitis	Wrong population
Smith, 2003	Mass antibiotic treatment for group A streptococcus outbreaks in two long-term care facilities	Wrong population
Spargen, 2011	Proinflammatory immune response and puerperal group a streptococcal sepsis	Wrong population
Steer, 2009	Prospective surveillance of invasive group A streptococcal disease, fiji, 2005-2007	Wrong population
Steer, 2008	High burden of invasive beta-haemolytic streptococcal infections in Fiji	Wrong population
Tanna, 2006	Molecular characterization of clinical isolates of M non-typable group A streptococci from invasive disease cases	Wrong population
Tapiainen, 2016	Invasive Group A Streptococcal Infections in Children: A Nationwide Survey in Finland	Wrong population
Thanert, 2019	Molecular profiling of tissue biopsies reveals unique signatures associated with streptococcal necrotizing soft tissue infections	Wrong population
Theis, 2002	Severe necrotising soft tissue infections in orthopaedic surgery	Wrong population
Thigpen, 2007	Nursing home outbreak of invasive group A streptococcal infections caused by 2 distinct strains	Wrong population
Urbina, 2019	Early identification of patients at high risk of group A streptococcus-associated necrotizing skin and soft tissue infections: a retrospective cohort study	Wrong population
Vasant, 2019	Mass prophylaxis in an outbreak of invasive group A streptococcal disease in a residential aged care facility	Wrong population
Vlaminckx, 2005	Long-term surveillance of invasive group A streptococcal disease in The Netherlands, 1994-2003	Wrong population
Vugia, 1996	Invasive group A streptococcal infections in children with varicella in Southern California	Wrong population
Waldhausen, 1996	Surgical implications of necrotizing fasciitis in children with chickenpox	Wrong population
Watanabe-Ohnishi, 1995	Selective depletion of V beta-bearing T cells in patients with severe invasive group A streptococcal infections and streptococcal toxic shock syndrome. Ontario Streptococcal Study Project	Wrong population
Wheeler, 1991	Outbreak of group A streptococcus septicemia in children: Clinical, epidemiologic, and microbiological correlates	Wrong population
Wilson, 1995	Group A streptococcal necrotizing fasciitis following varicella in children: Case reports and review	Wrong population
Wong, 2019	A Cluster of Pediatric Invasive Group A Streptococcus Disease in Melbourne, Australia, Coinciding with a High-Burden Influenza Season	Wrong population
Yagupsky, 1987	Group A beta-hemolytic streptococcal bacteremia in children	Wrong population
Zerr, 1999	A case-control study of necrotizing fasciitis during primary varicella	Wrong population
Zimbelman, 1999	Improved outcome of clindamycin compared with beta-lactam antibiotic treatment for invasive Streptococcus pyogenes infection	Wrong population

Abraham, 2016	Distribution of emm types of beta hemolytic streptococci associated with necrotizing fascitis: Clinical profile and outcome	Wrong population
Acosta, 2014	Severe Maternal Sepsis in the UK, 2011-2012: A National Case-Control Study	Wrong population
Adams, 2010	Investigation into an outbreak of invasive Group A Streptococcal (iGAS) infection at a general hospital in 2010	Wrong population
Adem, 2009	Identification of invasive streptococcal disease in a pediatric and infant death Cohort-Iowa, 2007-2008	Wrong population
Afifi, 2008	Acute necrotizing fasciitis in Egyptian patients: A case series	Wrong population
Al-Khadidi, 2017	Group A Streptococcal bacteraemia. Experience at King Fahad Medical City in Riyadh, Saudi Arabia	Wrong population
Alva, 2013	Necrotising fasciitis: A series of seven cases	Wrong population
Anonymous, 2007	Summaries for patients. Invasive streptococcal infections in hospitals in Ontario, Canada, 1992 to 2000	Wrong population
Aronoff, 2008	Postpartum invasive group A streptococcal disease in the modern era	Wrong population
Babbar, 2018	Pivotal Role of Preexisting Pathogen-Specific Antibodies in the Development of Necrotizing Soft-Tissue Infections	Wrong population
Babbar, 2016	A serological evaluation of the host immune response during Necrotizing Soft Tissue Infections caused by Streptococcus pyogenes	Wrong population
Babiker, 2019	Impact of adjunctive clindamycin in invasive beta-hemolytic streptococcal infections: A propensity score-matched analysis of 1956 beta-lactam treated patients from 118 us hospitals	Wrong population
Bajpai, 1977	Chemotherapy of acute bone and joint infections	Wrong population
Barnham, 2001	Bacteraemic Streptococcus pyogenes infection in the peri-partum period: now a rare disease and prior carriage by the patient may be important	Wrong population
Basma, 1999	Risk factors in the pathogenesis of invasive group A streptococcal infections: Role of protective humoral immunity	Wrong population
Bauer, 2015	Maternal deaths due to sepsis in the state of Michigan, 1999-2006	Wrong population
Beaudoin, 2014	Invasive group A Streptococcus infections associated with liposuction surgery at outpatient facilities not subject to state or federal regulation	Wrong population
Beigh, 2012	Postoperative complications followed by septoplasty comparison between conventional nasal packing and glove finger pack	Wrong population
Berkley, 1987	The relationship of tampon characteristics to menstrual toxic shock syndrome	Wrong population
Bingol-Kologlu, 2007	Necrotizing fasciitis in children: diagnostic and therapeutic aspects	Wrong population
Bruun, 2013	Necrotizing soft tissue infections caused by Streptococcus pyogenes and Streptococcus dysgalactiae subsp. equisimilis of groups C and G in western Norway	Wrong population
Bruun, 2020	Risk factors and Predictors of Mortality in Streptococcal Necrotizing Soft-Tissue Infections: A Multicenter Prospective Study	Wrong population

Busowski, 2013	Puerperal group a streptococcal infections: A case series and discussion	Wrong population
Byer, 2006	Clinical deterioration among patients with fever and erythroderma	Wrong population
Centers for Disease, 1982	Toxic-shock syndrome, United States, 1970-1982	Wrong population
Centers for Disease, 2011	Invasive group A streptococcus in a skilled nursing facility--Pennsylvania, 2009-2010	Wrong population
Chan, 2009	Toxic shock syndrome and rhinosinusitis in children	Wrong population
Chen, 2011	The microbiological profile and presence of bloodstream infection influence mortality rates in necrotizing fasciitis	Wrong population
Chen, 2015	Clinical Characteristics and Risk Factor Analysis for Lower-Extremity Amputations in Diabetic Patients With Foot Ulcer Complicated by Necrotizing Fasciitis	Wrong population
Chen, 2018	Macro- and Microvascular Parameters After Toxic Shock Syndrome	Wrong population
Ching, 2019	Prospective surveillance of pediatric invasive group A Streptococcus infection	Wrong population
Chiobotaru, 1997	Changing epidemiology of invasive Streptococcus pyogenes infections in southern Israel: differences between two ethnic population groups	Wrong population
Christie, 1988	Bacteremia with group A streptococci in childhood	Wrong population
Corona, 2016	Necrotising fasciitis of the extremities: implementation of new management technologies	Wrong population
Daneman, 2007	Surveillance for hospital outbreaks of invasive group a streptococcal infections in Ontario, Canada, 1992 to 2000	Wrong population
Daneman, 2005	Hospital-acquired invasive group A streptococcal infections in Ontario, Canada, 1992-2000	Wrong population
Davies, 1996	Invasive group A streptococcal infections in Ontario, Canada. Ontario Group A Streptococcal Study Group	Wrong population
Davis, 1982	Toxic shock syndrome: a critique of the 1980 Wisconsin case-control study	Wrong population
De Almeida Torres, 2013	Group a streptococcus meningitis in children	Wrong population
Deutscher, 2011	Incidence and severity of invasive Streptococcus pneumoniae, group A Streptococcus, and group B Streptococcus infections among pregnant and postpartum women	Wrong population
Devaney, 2015	Necrotising soft tissue infections: The effect of hyperbaric oxygen on mortality	Wrong population
Dooling, 2013	Investigation of a prolonged Group A Streptococcal outbreak among residents of a skilled nursing facility, Georgia, 2009-2012	Wrong population
Dworkin, 2009	The epidemiology of necrotizing fasciitis including factors associated with death and amputation	Wrong population
Eneli, 2007	Epidemiology and Outcome of Necrotizing Fasciitis in Children: An Active Surveillance Study of the Canadian Paediatric Surveillance Program	Wrong population
Factor, 2005	Risk factors for pediatric invasive group A streptococcal disease	Wrong population
Factor, 2003	Invasive group a streptococcal disease: Risk factors for adults	Wrong population

Fauno Thrane, 2017	Hyperbaric oxygen may only be optional in head and neck necrotizing fasciitis: a retrospective analysis of 43 cases and review of the literature	Wrong population
Fichtenbaum, 1993	Ehrlichiosis presenting as a life-threatening illness with features of the toxic shock syndrome	Wrong population
Flavahan, 2014	Incidence of periorbital necrotising fasciitis in the UK population: A BOSU study	Wrong population
Flores, 2019	Capsule-negative EMM types are an increasing cause of pediatric group a streptococcal infections at a large pediatric hospital in Texas	Wrong population
Frere, 2016	Clinical and Microbiological Characteristics of Invasive Group A Streptococcal Infections Before and After Implementation of a Universal Varicella Vaccine Program	Wrong population
Givner, 1998	Invasive disease due to group A beta-hemolytic streptococci: Continued occurrence in children in North Carolina	Wrong population
Givner, 1991	Apparent increase in the incidence of invasive group A beta-hemolytic streptococcal disease in children	Wrong population
Gooskens, 2005	Streptococcal toxic shock syndrome by an iMLS resistant M type 77 Streptococcus pyogenes in the Netherlands	Wrong population
Gonzalez, 1996	Necrotizing fasciitis of the upper extremity	Wrong population
Lesko, 2001	Invasive group A streptococcal infection and nonsteroidal antiinflammatory drug use among children with primary varicella	Wrong population
Lin, 2013	Group a streptococcal necrotizing fasciitis in the emergency department	Wrong population
Lin, 2015	Early Differentiation of Kawasaki Disease Shock Syndrome and Toxic Shock Syndrome in a Pediatric Intensive Care Unit	Wrong population
Parks, 2019	Elevated risk of invasive group A streptococcal disease and host genetic variation in the human leucocyte antigen locus	Wrong population
Stromberg, 1991	Outbreak of group A streptococcal bacteremia in Sweden: an epidemiologic and clinical study	Wrong population
Haggar, 2012	Clinical and microbiologic characteristics of invasive Streptococcus pyogenes infections in north and south India	Relationship between prognostic factor and outcome not reported or quantified
Laupland, 2000	Invasive group A streptococcal disease in children and association with varicella-zoster virus infection. Ontario Group A Streptococcal Study Group	Relationship between prognostic factor and outcome not reported or quantified
Linnemann, 1986	Increasing incidence of toxic shock syndrome in the 1970s	Relationship between prognostic factor and outcome not reported or quantified
Miday, 1988	Toxic shock syndrome: incidence and geographic distribution from a hospital medical records reporting system	Relationship between prognostic factor and outcome not reported or quantified
Mosites, 2018	Outbreak of Invasive Infections from Subtype emm26.3 Group A Streptococcus among Homeless Adults-Anchorage, Alaska, 2016-2017	Relationship between prognostic factor and outcome not reported or quantified

O'Grady, 2007	The epidemiology of invasive group A streptococcal disease in Victoria, Australia	Relationship between prognostic factor and outcome not reported or quantified
Petitti, 1989	Update through 1985 on the incidence of toxic shock syndrome among members of a prepaid health plan	Relationship between prognostic factor and outcome not reported or quantified
Pilon, 2019	Invasive group A streptococcal infection outbreaks of typeemm118 in a long-term care facility, and of type emm74 in the homeless population, Montreal, Quebec	Relationship between prognostic factor and outcome not reported or quantified
Rantala, 2012	Streptococcus pyogenes bacteraemia, emm types and superantigen profiles	Relationship between prognostic factor and outcome not reported or quantified
Tanner, 1981	Toxic shock syndrome	Relationship between prognostic factor and outcome not reported or quantified
Teatero, 2018	Canada-Wide Epidemic of emm74 Group A Streptococcus Invasive Disease	Relationship between prognostic factor and outcome not reported or quantified
Todd, 1985	Toxic shock syndrome. II. Estimated occurrence in Colorado as influenced by case ascertainment methods	Relationship between prognostic factor and outcome not reported or quantified
Tsai, 2014	Correlation of virulence genes to clinical manifestations and outcome in patients with Streptococcus dysgalactiae subspecies equisimilis bacteremia	Relationship between prognostic factor and outcome not reported or quantified
Vallalta Morales, 2006	Group A streptococcal bacteremia: outcome and prognostic factors	Relationship between prognostic factor and outcome not reported or quantified
Vlaminckx, 2004	Epidemiological features of invasive and noninvasive group A streptococcal disease in the Netherlands, 1992-1996	Relationship between prognostic factor and outcome not reported or quantified
Aydin, 2017	Clinical indications of intravenous immunoglobulin use in pediatric infectious diseases clinic	Relationship between prognostic factor and outcome not reported or quantified
Ben-Abraham, 2002	Invasive group A streptococcal infections in a large tertiary center: epidemiology, characteristics and outcome	Relationship between prognostic factor and outcome not reported or quantified
Bochicchio, 2001	Group A Streptococcus (GAS) soft-tissue infections: a lethal organism on the rise	Relationship between prognostic factor and outcome not reported or quantified
Cancellara, 2016	Multicenter study on invasive Streptococcus pyogenes infections in children in Argentina	Relationship between prognostic factor and

		outcome not reported or quantified
Chen, 2016	Toxic shock syndrome in Australian children	Relationship between prognostic factor and outcome not reported or quantified
Doctor, 1995	Group A beta-hemolytic streptococcal bacteremia: historical overview, changing incidence, and recent association with varicella	Relationship between prognostic factor and outcome not reported or quantified
Eriksson, 2003	Group A streptococcal infections in Sweden: a comparative study of invasive and noninvasive infections and analysis of dominant T28 emm28 isolates	Relationship between prognostic factor and outcome not reported or quantified
Norrby-Teglund, 2003	The treatment of severe group a streptococcal infections	Relationship between prognostic factor and outcome not reported or quantified
Rodriguez-Nunez, 2011	Clinical characteristics of children with group A streptococcal toxic shock syndrome admitted to pediatric intensive care units	Relationship between prognostic factor and outcome not reported or quantified
Snall, 2016	Differential neutrophil responses to bacterial stimuli: Streptococcal strains are potent inducers of heparin-binding protein and resistin-release	Relationship between prognostic factor and outcome not reported or quantified
Eriksson, 1998	Epidemiological and clinical aspects of invasive group A streptococcal infections and the streptococcal toxic shock syndrome	Relationship between prognostic factor and outcome not reported or quantified
Sahli, 2014	Necrotizing fasciitisin in diabetic patients: A report of 14 cases	Not in English
Arnholm, 2004	High-dose immunoglobulin - Life-saving in invasive group a streptococcal infection	Not in English
Caetano, 2010	[S. Pyogenes invasive disease in a paediatric hospital: 1996-2009]	Not in English
Costa Orvay, 2007	[Toxic shock syndrome: experience in a pediatric intensive care unit]	Not in English
Dosil Gallardo, 2009	[Streptococcal toxic shock syndrome: an emerging pathology?]	Not in English
Emmi, 1999	Severe infection from invasive beta-hemolytic streptococcus group A. Three cases of toxic shock observed in resuscitation	Not in English
Faye, 2014	Management of severe invasive group A streptococcal infections	Not in English
Floret, 2001	Clinical aspects of staphylococcal and streptococcal toxinic diseases	Not in English
Hua, 2018	[Streptococcal toxic shock syndrome caused by Streptococcus pyogenes: a retrospective study of 15 pediatric cases]	Not in English
Kach, 1993	[Necrotizing soft tissue infections]	Not in English
Kaul, 1999	Intravenous immunoglobulin therapy for streptococcal toxic shock syndrome--a comparative observational study. The Canadian Streptococcal Study Group	Duplicate
Salinas, 2005	Immunoglobulins in sepsis and septic shock	Not in English

Shands, 1982	Toxic shock syndrome: case-control studies at the Centers for Disease Control	Duplicate
Sierra, 2006	Group A streptococcal infections in injection drug users in Barcelona, Spain: epidemiologic, clinical, and microbiologic analysis of 3 clusters of cases from 2000 to 2003	Duplicate
Urbina, 2019	Early identification of patients at high risk of group A streptococcus-associated necrotizing skin and soft tissue infections: A retrospective cohort study	Duplicate
Vallalta-Morales, 2005	[Streptococcal toxic shock syndrome: ten years' experience at a tertiary hospital]	Not in English
Vasant, 2019	Mass prophylaxis in an outbreak of invasive group A streptococcal disease in a residential aged care facility	Duplicate
Vugia, 1996	Invasive group A streptococcal infections in children with varicella in Southern California	Duplicate
Zachariadou, 2014	Differences in the epidemiology between paediatric and adult invasive Streptococcus pyogenes infections	Duplicate
Bernard, 1995	[Bacterial dermo-hypodermatitis in adults. Incidence and role of streptococcal etiology]	Not in English
Bleton, 1991	Necrotizing fasciitis of the upper limb. 12 cases	Not in English
Bleton, 1991	Necrotising fasciitis of the upper limb. Report of twelve cases	Duplicate
Fan, 2014	[Clinical characteristics and antimicrobial resistance of invasive group A beta-hemolytic streptococcus infection in children]	Not in English
Fica, 2003	Molecular epidemiology of a streptococcus pyogenes related nosocomial outbreak in a burn unit	Not in English
Fredlund, 1990	[Increased frequency of group A hemolytic streptococci. Old age and immune deficiency among the risk factors]	Not in English
Georgiev, 1993	Puerperal streptococcal sepsis	Not in English
Khan, 2003	Treatment of necrotizing fasciitis with quinolones	Wrong population
Frosi, 1999	Necrotizing fasciitis: A serious and uncommon alcohol related disease	Wrong study design
Nedrebo, 2020	Necrotizing Soft Tissue Infections: Case Reports, from the Clinician's Perspectives.	Wrong study design
Reicher, 2021	450 Obstetrically related group a streptococcal infection and predictors for severity - retrospective 12-year cohort study	Wrong study design
Bajpai, 2020	Anti-microbial-resistance and profile of exotoxins of invasive beta-haemolytic-streptococci infections in trauma patients	Wrong population
Adamkova, 2020	Can gram-negative-like biomarker values in Streptococcus pyogenes sepsis negatively influence right choice of initial antibiotic therapy?	Wrong population
Bandi, 2021	Group A streptococcal infections in children	Wrong population
Ceccato, 2020	Use of corticosteroids in patients with severe CAP admitted to ICU, experience in a real-life setting	Wrong population
Tepper, 2021	Long-term safety and tolerability of atogepant following once daily dosing over 1 year for the preventive treatment of migraine	Wrong population

Melo, 2021	Clinical, epidemiological and microbiological features of streptococcus pyogenes invasive infection - 10 year retrospective review	Wrong population
Bringel, 2021	Clinical characteristics and outcomes of children with toxic shock syndrome admitted to a pediatric intensive care unit: A case series	Wrong population
Neff, 2020	Characterisation of clinical manifestations and treatment strategies for invasive beta-haemolytic streptococcal infections in a Swiss tertiary hospital.	Wrong population
Urbina, 2020	Assessing and applying individualized treatment for group A streptococcal necrotizing soft-tissue infection is possible	Wrong population
Bergsten, 2020	Correlation between immunoglobulin dose administered and plasma neutralization of streptococcal superantigens in patients with necrotizing soft tissue infections	Wrong population
Boukthir, 2020	A prospective survey of Streptococcus pyogenes infections in French Brittany from 2009 to 2017: Comprehensive dynamic of new emergent emm genotypes.	Wrong population
Escrihuela-Vidal, 2021	Clinical Features and Outcomes of Streptococcus anginosus Group Infective Endocarditis: A Multicenter Matched Cohort Study.	Wrong population
Babiker, 2021	Effectiveness of adjunctive clindamycin in beta-lactam antibiotic-treated patients with invasive beta-haemolytic streptococcal infections in US hospitals: a retrospective multicentre cohort study.	Wrong population
Cui, 2021	Necrotizing soft tissue infection: clinical characteristics, diagnosis, and management of 32 cases in Beijing.	Wrong population
Link-Gelles, 2020	Characteristics of Intracranial Group A Streptococcal Infections in US Children, 1997-2014.	Wrong population
Peetermans, 2020	Use of Intravenous Immunoglobulins in Patients with Suspected Toxin-Mediated Shock Requiring Extracorporeal Membrane Oxygenation.	Wrong population
Bruun, 2020	Beta-Hemolytic Streptococci and Necrotizing Soft Tissue Infections.	Wrong population
Lima-Setta, 2021	Multisystem inflammatory syndrome in children (MIS-C) during SARS-CoV-2 pandemic in Brazil: a multicenter, prospective cohort study.	Wrong population
Kohler, 2020	Kininogen supports inflammation and bacterial spreading during Streptococccus Pyogenes Sepsis.	Wrong population
Bruun, 2021	Risk Factors and Predictors of Mortality in Streptococcal Necrotizing Soft-tissue Infections: A Multicenter Prospective Study.	Wrong population
Bjorck, 2020	Morbidity and mortality in critically ill patients with invasive group A streptococcus infection: an observational study.	Wrong population
Contou, 2021	Menstrual toxic shock syndrome: a French nationwide multicenter retrospective study.	Wrong population
Billon, 2020	Association of characteristics of tampon use with menstrual toxic shock syndrome in France.	Wrong population
Canetti, 2021	Invasive Group A Streptococcus Infection in Children in Central Israel in 2012-2019	Relationship between prognostic factor and outcome not reported or quantified

Vilhonen, 2020	Group A streptococcal bacteremias in Southwest Finland 2007-2018: epidemiology and role of infectious diseases consultation in antibiotic treatment selection.	Relationship between prognostic factor and outcome not reported or quantified
Thielemans, 2020	Clinical Description and Outcomes of Australian Children With Invasive Group A Streptococcal Disease.	Relationship between prognostic factor and outcome not reported or quantified
Madsen, 2020	Treatment of Necrotizing Soft Tissue Infections: IVIG	Wrong study design
Deniskin, 2019	Clinical Manifestations and Bacterial Genomic Analysis of Group A Streptococcus Strains That Cause Pediatric Toxic Shock Syndrome.	Relationship between prognostic factor and outcome not reported or quantified
Mosites, 2018	Outbreak of Invasive Infections From Subtype emm26.3 Group A Streptococcus Among Homeless Adults-Anchorage, Alaska, 2016-2017.	Relationship between prognostic factor and outcome not reported or quantified
Hasin, 2020	Invasive Group A Streptococcus Infection in Children in Southern Israel Before and After the Introduction of Varicella Vaccine.	Wrong population
Ching, 2019	Prospective Surveillance of Pediatric Invasive Group A Streptococcus Infection.	Wrong population
Loewen, 2017	Epidemiologic features of invasive group A Streptococcus infection in a rural hospital: 6-year retrospective report and literature review.	Wrong population
Bergsten, 2020	Correlation Between Immunoglobulin Dose Administered and Plasma Neutralization of Streptococcal Superantigens in Patients With Necrotizing Soft Tissue Infections.	Wrong population
Kobayashi, 2016	A Cluster of Group A Streptococcal Infections in a Skilled Nursing Facility-the Potential Role of Healthcare Worker Presenteeism.	Wrong population
Adebanjo, 2020	Evaluating Household Transmission of Invasive Group A Streptococcus Disease in the United States Using Population-based Surveillance Data, 2013-2016.	Wrong population
Link-Gelles, 2020	Characteristics of Intracranial Group A Streptococcal Infections in US Children, 1997-2014.	Wrong population
Bruun, 2021	Risk Factors and Predictors of Mortality in Streptococcal Necrotizing Soft-tissue Infections: A Multicenter Prospective Study.	Wrong population
Shah, 2015	Role of intravenous immune globulin in streptococcal toxic shock syndrome and Clostridium difficile infection.	Wrong study design
Hayata, 2021	Nationwide study of mortality and survival in pregnancy-related streptococcal toxic shock syndrome.	Duplicate
Fernandez-Galilea, 2022	Clindamycin but not Intravenous Immunoglobulins reduces mortality in a retrospective cohort of critically ill patients with bacteremic Group A Streptococcal infections.	Wrong population
Heil, 2021	Role of Clindamycin Versus Linezolid for Serious Group A Streptococcal Infections	Wrong population
Nanduri, 2022	Challenges in Surveillance for Streptococcal Toxic Shock Syndrome: Active Bacterial Core Surveillance, United States, 2014-2017.	Wrong study design

Hamada, 2022	Association between adjunct clindamycin and in-hospital mortality in patients with necrotizing soft tissue infection due to group A Streptococcus: a nationwide cohort study.	Wrong population
Fay, 2021	Patterns of Antibiotic Nonsusceptibility Among Invasive Group A Streptococcus Infections,ÄîUnited States, 2006,Äì2017.	Relationship between prognostic factor and outcome not reported or quantified
Horn, 2021	Outcomes of Œ≤-Hemolytic Streptococcal Necrotizing Skin and Soft-tissue Infections and the Impact of Clindamycin Resistance.	Relationship between prognostic factor and outcome not reported or quantified
Contou, 2022	Menstrual Toxic Shock Syndrome: A French Nationwide Multicenter Retrospective Study.	Wrong population
Valenciano, 2021	Invasive Group A Streptococcal Infections Among People Who Inject Drugs and People Experiencing Homelessness in the United States, 2010,Äì2017.	Relationship between prognostic factor and outcome not reported or quantified
Jutras, 2021	Intravenous Immunoglobulin Use In Critically Ill Children.	Wrong population
Metcalf, 2022	Cluster Transmission Drives Invasive Group A Streptococcus Disease Within the United States and Is Focused on Communities Experiencing Disadvantage.	Wrong population
Dunne, 2022	Increasing Incidence of Invasive Group A Streptococcus Disease, Idaho, USA, 2008-2019.	Relationship between prognostic factor and outcome not reported or quantified
VanZeeland, 2022	Public health response following an iGAS outbreak in a residential aged care facility in Queensland.	Wrong population
Barisiene, 2021	Lithuanian tertiary pediatric centre experience of multi-system inflammatory syndrome in children (MIS-C): clinical cases study	Wrong population
Silvestre, 2022	Toxic shock syndrome: diagnosis and management.	Wrong study design
Nabarro, 2022	Invasive Group A Streptococcus Outbreaks Associated with Home Healthcare, England, 2018-2019.	Wrong population
Nagata, 2022	Necrotizing fasciitis of the extremities in high and low Charlson Comorbidity Index: A multi-center retrospective cohort study.	Wrong population
deNeergaard, 2022	Invasive streptococcal infection can lead to the generation of cross-strain opsonic antibodies	Wrong population
Pershing, 2021	Pediatric Group A Streptococcal Peritonitis: A Single-Center Eleven Patient Case Series	Wrong population
Nawijn, 2021	Incidence and mortality of necrotizing fasciitis in The Netherlands: the impact of group A Streptococcus.	Wrong population
Sahin, 2022	Clinical and Laboratory Features of Invasive Group A Streptococcal Infections: 8 Years Experience.	Wrong population
Thean, 2020	The epidemiology and clinical course of invasive staphylococcus aureus and group a streptococcus infections in Fiji: A prospective study	Wrong population

Hayata, 2021	Nationwide study of mortality and survival in pregnancy-related streptococcal toxic shock syndrome	Duplicate

Table of additional study characteristics

Study	Study Design	Country	Number of STSS cases	Mean age	% Male	Cardiovascular Disease (%)	Diabetes (%)	Presence of Necrotizing Fasciitis (%)	Probable GAS STSS (%)	Confirmed GAS STSS (%)	Prognostic factor and outcome combination of interest reported
Abuhammour 2004	Cohort	United States	2	9	100	NR	NR	NR	NR	NR	age - clinical cure/improvement^ age - ICU admission^ age - mortality^ any antibiotic - clinical cure/improvement^ any antibiotic - ICU admission any antibiotic - mortality
Adalat 2014	Cohort	United Kingdom, Ireland	29	4 (median)	62	NR	NR	28	38	62	IVIG - mortality
Al-Ajmi 2012	Case-series	Qatar	2	35	0	NR	NR	NR	0	100	age - clinical cure/improvement^ age - ICU admission^ age - mortality^
Barnham 2002	Case-series	England	12	57	64	NR	NR	58	17	83	age - ICU admission^ age - mortality^ any antibiotic - ICU admission any antibiotic - mortality clindamycin - ICU admission^ clindamycin - mortality emm type - ICU admission^ emm type - mortality^ immunocompromised - ICU admission^ immunocompromised - mortality IVIG - ICU admission IVIG - mortality IVIG - time to mortality^ NF - ICU admission NF - mortality NSAIDs - ICU admission NSAIDs - mortality
Bernaldo de Quiros 1997	Cohort	Spain	9	NR	NR	NR	NR	NR	NR	NR	immunocompromised - mortality^

Study	Study Design	Country	Number of STSS cases	Mean age	% Male	Cardiovascular Disease (%)	Diabetes (%)	Presence of Necrotizing Fasciitis (%)	Probable GAS STSS (%)	Confirmed GAS STSS (%)	Prognostic factor and outcome combination of interest reported
Brogan 1995	Case-series	United States	5	6	60	NR	NR	100	60	40	age - clinical cure/improvement^ age - hospital LOS^ age - ICU admission^ age - ICU LOS^ age - mortality^ NSAIDs - clinical cure/improvement^ NSAIDs - hospital LOS^ NSAIDS - ICU admission NSAIDs - ICU LOS^ NSAIDs - mortality sex - clinical cure/improvement sex - hospital LOS^ sex - ICU admission sex - ICU LOS^ sex - mortality
Butragueno Laiseca 2017	Case-series	Spain	13	2	50	NR	NR	15	15	85	acute renal failure - clinical cure/improvement^ acute renal failure - mechanical ventilation acute renal failure - mortality age - clinical cure/improvement^ age - ICU LOS^ age - mechanical ventilation^ age - mortality^ clindamycin - clinical cure/improvement^ clindamycin - ICU LOS^ clindamycin - mechanical ventilation^ clindamycin - mortality hemodialysis - clinical cure/imrpovement hemodialysis - mechanical ventilation hemodialysis - mortality IVIG - clinical cure/improvement IVIG - ICU LOS IVIG - mechanical ventilation IVIG - mortality NF - clinical cure/improvement NF - ICU LOS^ NF - mechanical ventilation NF - mortality
Carapetis 1995	Case-series	Australia	5	NR	NR	NR	NR	NR	NR	NR	age - mortality^

Study	Study Design	Country	Number of STSS cases	Mean age	% Male	Cardiovascular Disease (%)	Diabetes (%)	Presence of Necrotizing Fasciitis (%)	Probable GAS STSS (%)	Confirmed GAS STSS (%)	Prognostic factor and outcome combination of interest reported
Carapetis 2014	Cohort	Australia	37	60	50	NR	NR	100	0	100	age - mortality^ IVIG - mortality sex - mortality
Cimolai 1992	Case-series	Canada	4	8	50	NR	NR	NR	0	100	age - clinical cure/improvement^ age - mortality^ sex - clinical cure/improvement sex - mortality
Cook 2020	Cohort	United States	27	9	44	NR	NR	19	56	44	other - other^
Cowan 1994	Case-series	United States	3	2	67	NR	NR	NR	100	0	age - clinical cure/improvement^ age - ICU admission^ age - ICU LOS^ age - mechanical ventilation^ age - mortality^ sex - clinical cure/improvement sex - ICU admission sex - ICU LOS^ sex - mechanical ventilation sex - mortality
Crum 2004	Case-series	United States	2	24	100	NR	NR	50	0	100	age - clinical cure/improvement^ age - ICU admission^ age - mechanical ventilation^ age - mortality^ hemodialysis - clinical cure/improvement hemodialysis - ICU admission hemodialysis - mechanical ventilation hemodialysis - mortality
Dahl 2002	Case-series	United States	5	53	NR	NR	NR	100	0	100	age - mortality^ immunocompromised - mortality
Darenberg 2003	Randomized trial	Sweden, Norway, Finland, Netherlands	18	52	48	NR	NR	NR	11	89	IVIG - change in SOFA score^ IVIG - mortality IVIG - time to clinical cure/improvement^ IVIG - time to mortality^

Study	Study Design	Country	Number of STSS cases	Mean age	% Male	Cardiovascular Disease (%)	Diabetes (%)	Presence of Necrotizing Fasciitis (%)	Probable GAS STSS (%)	Confirmed GAS STSS (%)	Prognostic factor and outcome combination of interest reported
Donaldson 1993	Case-series	England	5	67	20	NR	NR	100	NR	NR	age - mortality^ any antibiotic - mortality sex - mortality
Erdem 2004	Case-series	United States	3	52	67	NR	NR	100	NR	33	age - mortality^ sex - mortality
Eriksson 1999	Cohort	Sweden	6	46	83	NR	NR	NR	0	100	age - mortality^ emm type - mortality^ sex - mortality
Forni 1995	Case-series	United States	5	54	83	NR	17	50	0	100	acure renal failure - ICU admission^ acute renal failure - mortality age - hospital LOS^ age - ICU admission^ age - mortality^ emm type - ICU admission^ emm type - mortality^ NF - ICU admission NF - mortality
Fronhoffs 2000	Case-series	Germany	7	49	71	14	14	86	86	14	acute renal failure - mechanical ventilation acute renal failure - mortality age - mechanical ventilation^ age - mortality^ immunocompromised - mechanical ventilation^ immunocompromised - mortality NF - mechanical ventilation NF - mortality NSAIDs - mechanical ventilation^ NSAIDs - mortality sex - mechanical ventilation sex - mortality
Hasegawa 2004	Case-control	Japan	66*	Range: 0 to 70	59	NR	18	NR	100	0	acute renal failure - mortality
Hayata 2021	Cohort	Japan	28	NR	0	NR	NR	NR	0	100	age - mortality^ NSAIDs - mortality

Study	Study Design	Country	Number of STSS cases	Mean age	% Male	Cardiovascular Disease (%)	Diabetes (%)	Presence of Necrotizing Fasciitis (%)	Probable GAS STSS (%)	Confirmed GAS STSS (%)	Prognostic factor and outcome combination of interest reported
Huang 2001	Case-series	Taiwan	3	6	100	NR	NR	NR	33	67	age - mortality^
Kansal 2000	Cohort	Canada	28	NR	NR	NR	NR	NR	NR	NR	NF - other^
Kaul 1999	Case-control	Canada	53	57	55	NR	NR	NR	NR	NR	clindamycin - mortality IVIG - duration of mechanical ventilation^ IVIG - hospital LOS IVIG - mortality NF - mortality
Linder 2017	Cohort	United States	10	NR	NR	NR	NR	NR	0	100	immunocompromised - mortality
Linnér 2014	Cohort	Sweden	67	63	42	48	16	28	NR	NR	age - mortality^ clindamycin - mortality IVIG - hospital LOS IVIG - mortality
Luca-Harari 2009	Cohort	Cyprus, Czech Republic, Denmark, Finand, France, Germany, Greece, Italy, Romania, Sweden, United Kingdom	476	NR	NR	NR	NR	NR	NR	NR	emm type - mortality^
Nelson 2016	Cohort	United States	381	NR	NR	NR	NR	19	NR	NR	age - mortality NF - mortality
O'Loughlin 2007	Cohort	United States	309	NR	NR	NR	NR	20	NR	NR	age - mortality NF - mortality
Page 2011	Cohort	Canada	16	NR	NR	NR	NR	NR	NR	NR	other - other^
Safar 2011	Cohort	New Zealand	30	NR	NR	NR	NR	17	0	100	age - mortality NF - mortality
Schwartz 1992	Case-series	United States	8	40	40	NR	NR	NR	0	100	age - mortality^ emm type - mortality^ sex - mortality
Shah 2009	Cohort	United States	192	8	49	NR	NR	NR	NR	NR	IVIG - cost^ IVIG - hospital LOS IVIG - ICU admission IVIG - ICU LOS IVIG - mechanical ventilation IVIG - mortality

Study	Study Design	Country	Number of STSS cases	Mean age	% Male	Cardiovascular Disease (%)	Diabetes (%)	Presence of Necrotizing Fasciitis (%)	Probable GAS STSS (%)	Confirmed GAS STSS (%)	Prognostic factor and outcome combination of interest reported
Sriskandan 2000	Cohort	England	7	NR	NR	NR	NR	NR	NR	NR	other - other^
Stegmayr 1992	Case-series	Sweden	11	42	64	NR	NR	NR	27	73	age - clinical cure/improvement^ age - ICU admission^ age - mechanical ventilation^ age - mortality^ hemodialysis - clinical cure/improvement hemodialysis - ICU admission hemodialysis - mechanical ventilation hemodialysis - mortality IVIG - clinical cure/improvement IVIG - ICU admission IVIG - mechanical ventilation IVIG - mortality NF - clinical cure/improvement NF - ICU admission NF - mechanical ventilation NF - mortality sex - clinical cure/improvement sex - ICU admission sex - mechanical ventilation sex - mortality
Stevens 1989	Case-series	United States	19	41	53	80	5	21	0	100	age - mortality^ emm type - mortality^ hemodialysis - mortality NF - mortality sex - mortality
Stockmann 2012	Cohort	United States	53	30 (median)	NR	58	19	32	NR	NR	age - ICU admission^ age - mortality
Tagini 2017	Case-series	Switzerland	5	18	60	NR	NR	NR	20	80	age - mortality^ emm type - mortality^ sex - mortality
Torimitsu 2021	Case-series	Japan	4	NR	75	NR	50	25	0	100	sex - mortality
Zachariadou 2013	Cohort	Greece	19	NR	NR	NR	NR	NR	0	100	age - mortality emm type - mortality^

*More than 80% of STSS cases due to group A *Streptococcus*
^Excluded from meta-analysis

NF=necrotizing fasciitis

NSAIDs=non-steroidal anti-inflammatory drugs
ICU=intensive care unit
IVIG=intravenous immunoglobulin
GAS=group A *Streptococcus*
STSS=streptococcal toxic shock syndrome
NR=not reported

Risk of bias assessment of included studies

Author	Study participation	Study Attrition	Prognostic Factor Measurement	Outcome Measurement	Study Confounding	Statistical Analysis and Presentation Summary	Overall Risk of Bias
Abuhammour 2004	Low	Low	Low	Low	High	High	High
Adalat 2014	Low	High	High	Low	High	High	High
Al-Ajmi 2012	NA	NA	NA	NA	NA	NA	High
Barnham 2002	NA	NA	NA	NA	NA	NA	High
Bernaldo de Quiros 1997	Low	Low	Low	Low	High	High	High
Brogan 1995	NA	NA	NA	NA	NA	NA	High
Butragueno Laiseca 2017	NA	NA	NA	NA	NA	NA	High
Carapetis 1995	NA	NA	NA	NA	NA	NA	High
Carapetis 2014	Low	Low	Low	Low	High	High	High
Cimolai 1992	NA	NA	NA	NA	NA	NA	High
Cook 2021	Low	Low	Low	Low	High	High	High
Cowan 1994	NA	NA	NA	NA	NA	NA	High
Crum 2004	NA	NA	NA	NA	NA	NA	High
Dahl 2002	NA	NA	NA	NA	NA	NA	High
Darenberg 2003	Low	Moderate	Low	Moderate	Low	Low	Moderate
Donaldson 1993	NA	NA	NA	NA	NA	NA	High
Erdem 2004	NA	NA	NA	NA	NA	NA	High
Eriksson 1999	Low	Low	Low	Low	High	High	High
Forni 1995	NA	NA	NA	NA	NA	NA	High
Fronhoffs 2000	NA	NA	NA	NA	NA	NA	High
Hasegawa 2004	Low	Low	Moderate	Low	High	High	High
Hayata 2021	Low	Low	Low	Low	High	High	High
Huang 2001	NA	NA	NA	NA	NA	NA	High
Kansal 2000	Moderate	Low	Moderate	Low	High	High	High
Kaul 1999	Low	Low	Low	Low	High	High	High

Linder 2017	Low	Moderate	Low	Moderate	High	High	High
Linner 2014	Low	Low	Low	Low	Moderate	Low	Low
Luca-Harari 2009	Low	High	Low	Low	High	High	High
Nelson 2016	Low	Low	Low	Low	High	High	High
O'Loughlin 2007	Low	Low	Low	Low	High	High	High
Page 2011	Low	Low	Low	Low	Moderate	High	Moderate
Safar 2011	Low	Low	Low	Low	High	High	High
Schwartz 1992	NA	NA	NA	NA	NA	NA	High
Shah 2009	Low	High	Low	Low	Moderate	Moderate	Moderate
Sriskandan 2000	Moderate	Moderate	High	Low	High	High	High
Stegmayr 1992	NA	NA	NA	NA	NA	NA	High
Stevens 1989	NA	NA	NA	NA	NA	NA	High
Stockmann 2012	Low	Low	Low	Low	High	High	High
Tagini 2017	NA	NA	NA	NA	NA	NA	High
Torimitsu 2021	NA	NA	NA	NA	NA	NA	High
Zachariadou 2013	Low	Low	Low	Low	High	High	High

NA, Not Applicable because case-series study design and rated at high risk of bias.

Forest plots

n_e: number of patients with the outcome exposed to or experiencing the prognostic factor (experimental group)
N_e: total number of patients exposed to or experiencing the prognostic factor (experimental group)
n_c: number of patients with the outcome not exposed to or experiencing the prognostic factor (control group)
N_c: total number of patients not exposed to or experiencing the prognostic factor (control group)

Percentages in forest plots correspond to the percent weight contribution of each study in the meta-analysis.

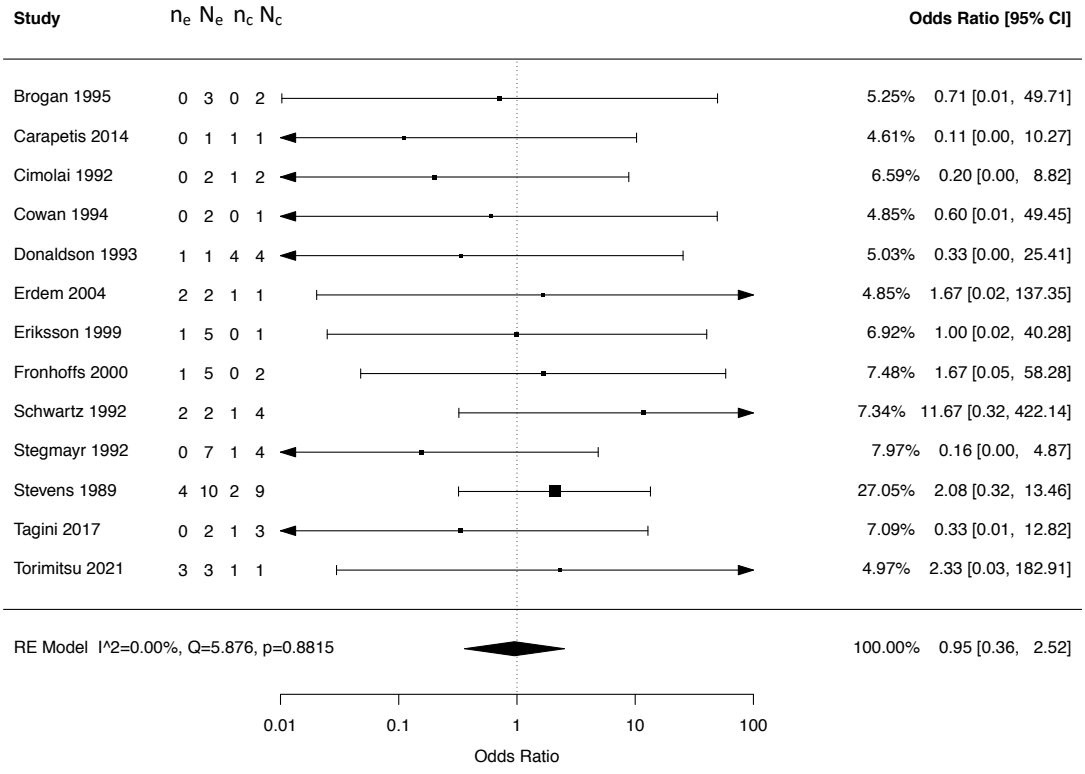
For mortality, ICU admission and need for mechanical ventilation outcomes, an odds ratio greater than 1 corresponds with a worse STSS prognosis and an odds ratio less than 1 corresponds with a better STSS prognosis.

For clinical cure or improvement outcome, an odds ratio greater than 1 corresponds with a better STSS prognosis and an odds ratio less than 1 corresponds with a worse STSS prognosis.

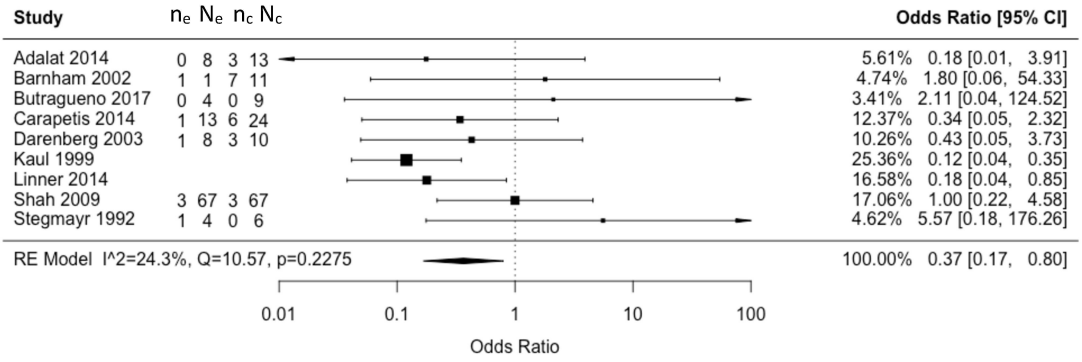
For duration of hospitalization and ICU stay outcomes, a mean difference greater than 1 corresponds with a worse STSS prognosis and a mean difference less than 1 corresponds with a better STSS prognosis.

Mortality

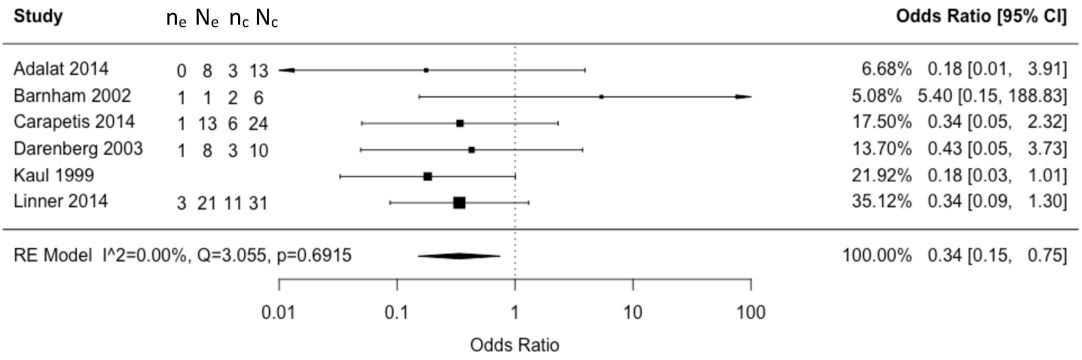
1. Sex: male vs female (reference)



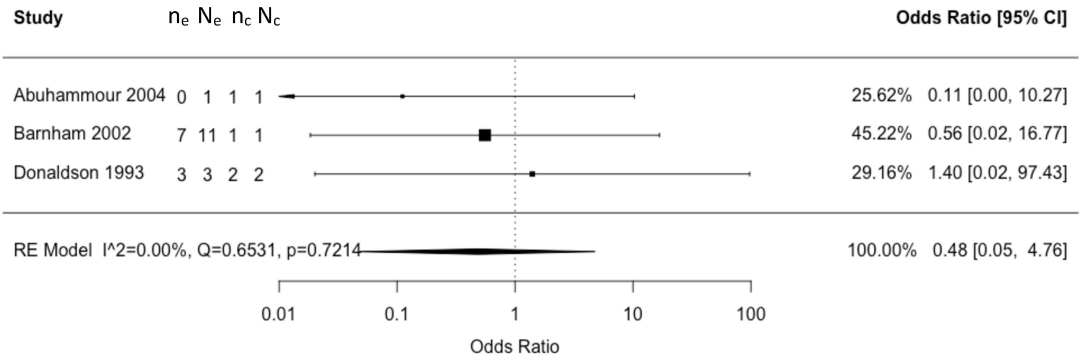
2.A) IVIG in all STSS patients: yes vs no (reference)

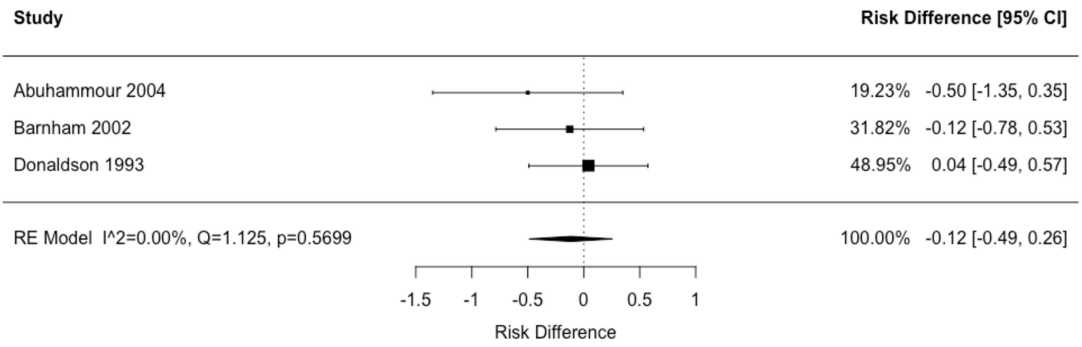


2.B) IVIG in the subset of STSS patients treated with clindamycin: yes vs no (reference)

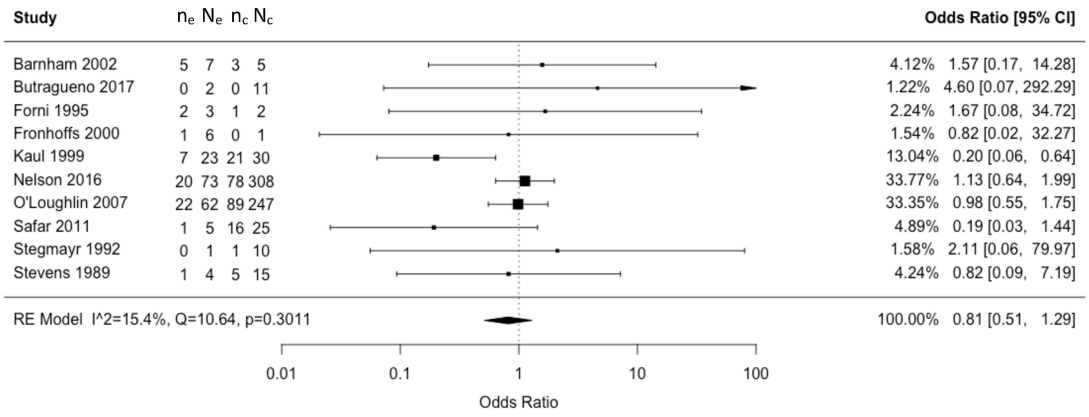


3. Any antibiotic: yes vs no (reference)

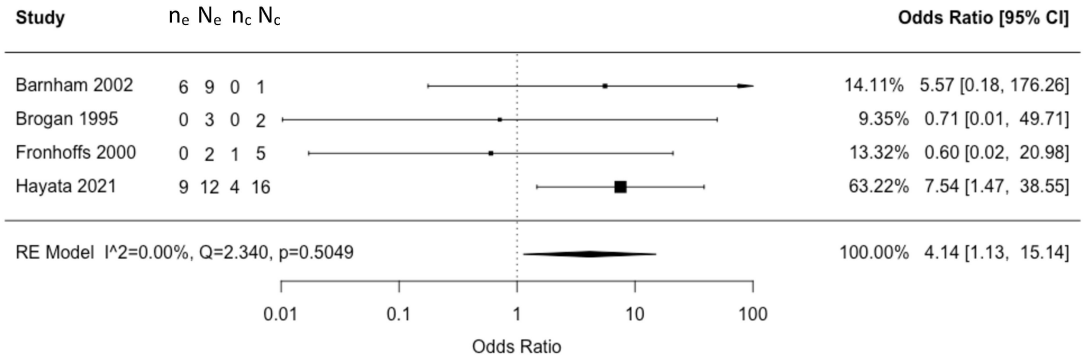




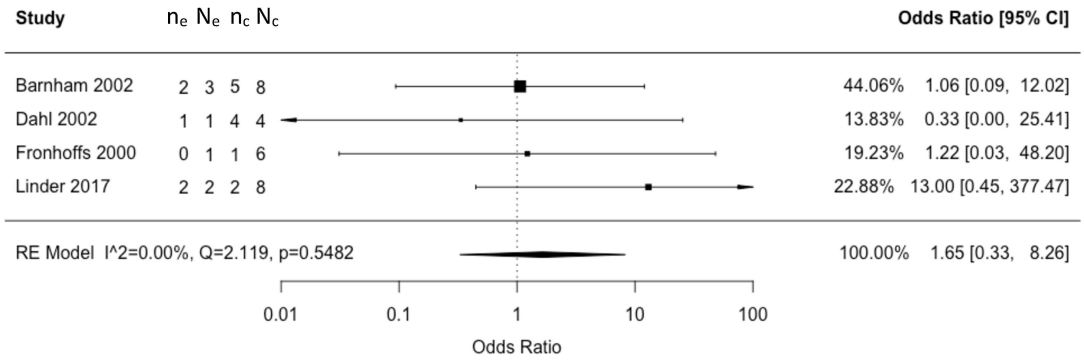
4. Necrotizing fasciitis: yes vs no (reference)



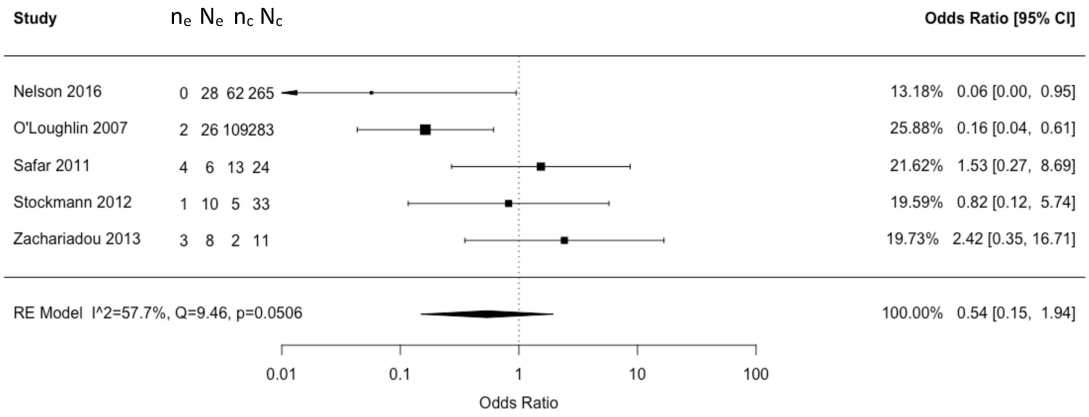
5. NSAIDs: yes vs no (reference)



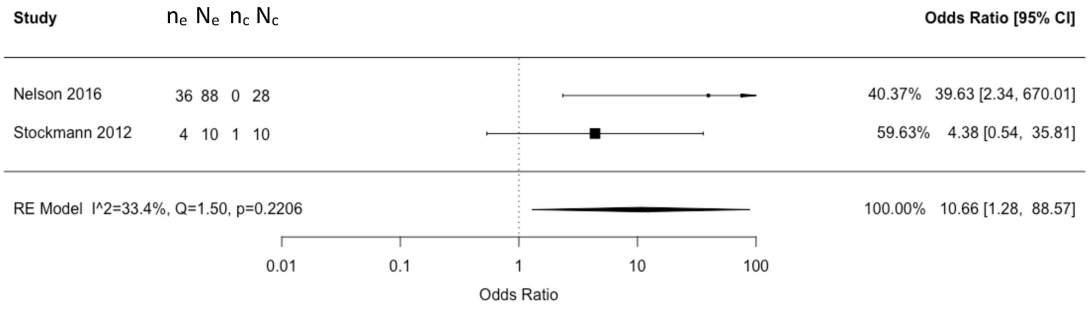
6. Immunocompromised: yes vs no (reference)



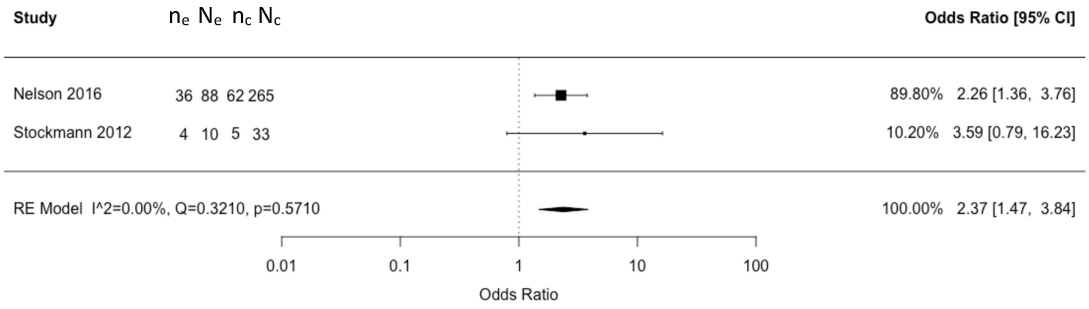
7. Age: <18 years vs 18-64 years (reference)



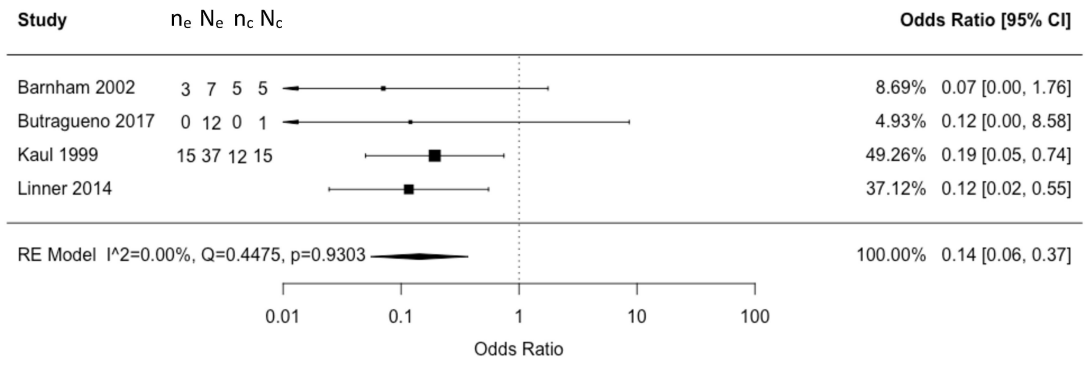
8. Age: ≥65 years vs <18 years (reference)



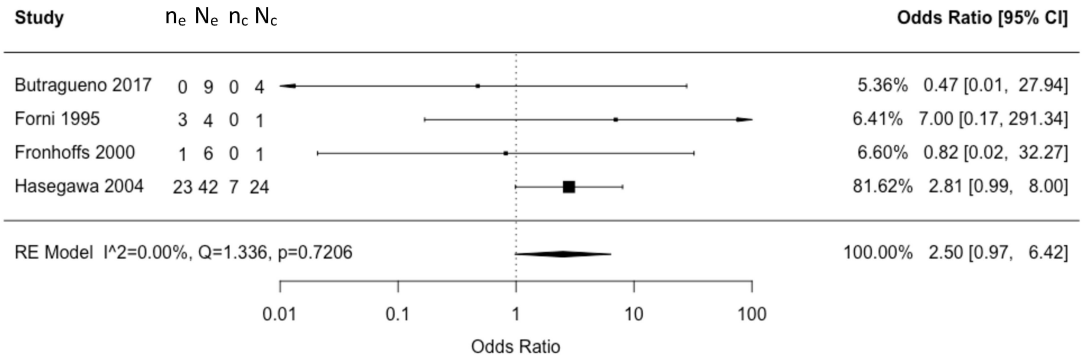
9: Age: ≥65 years vs 18-64 years (reference)

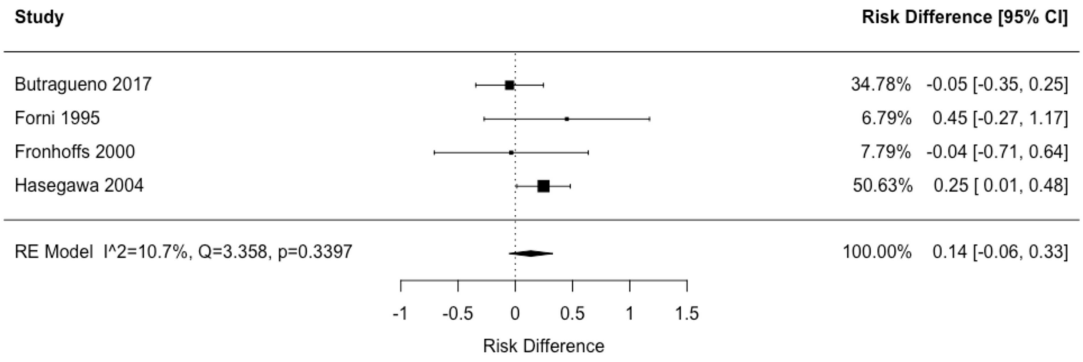


10. Clindamycin antibiotic vs no clindamycin antibiotic (reference)

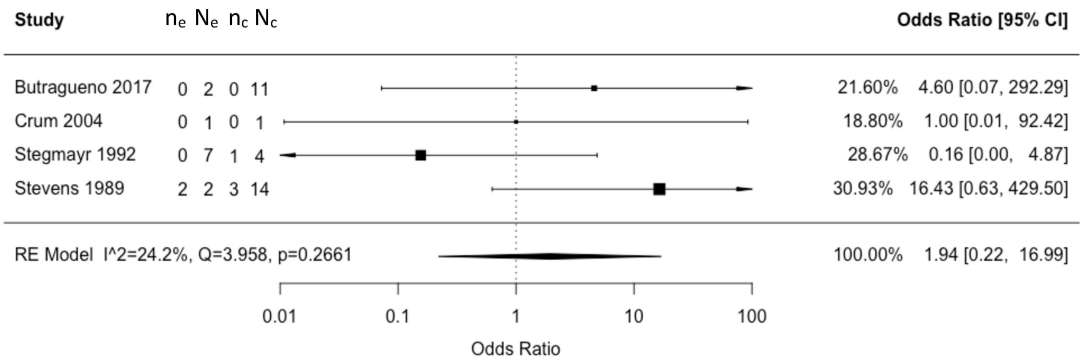


11. Acute renal failure: yes vs no (reference)





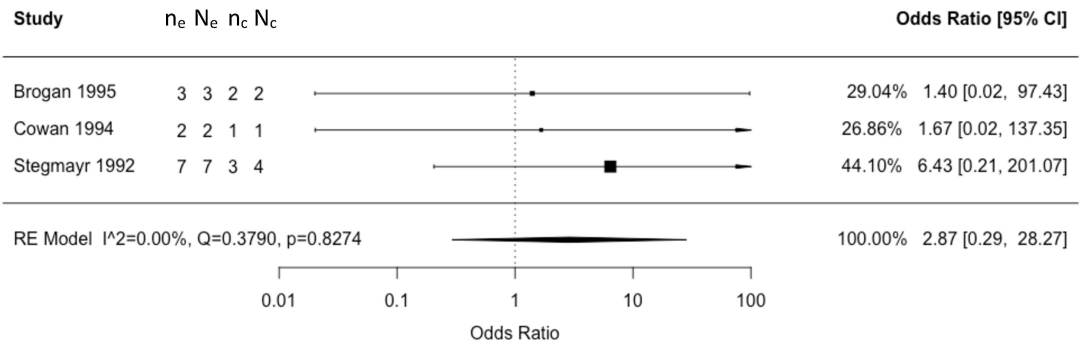
12. Hemodialysis: yes vs no (reference)

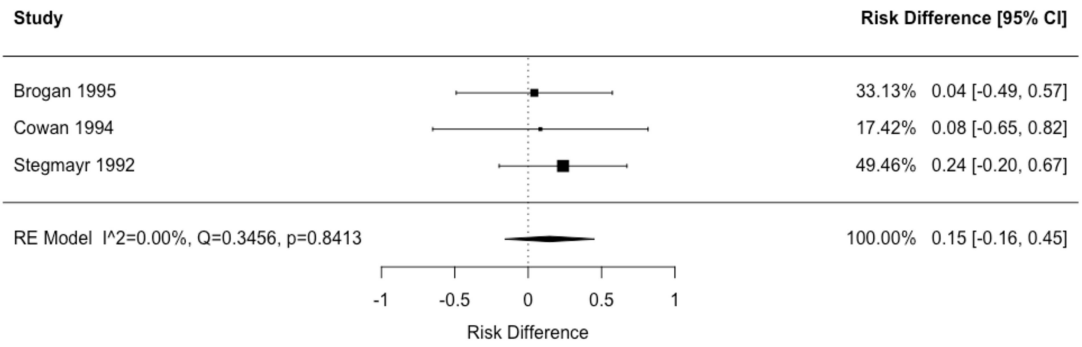


ICU admission

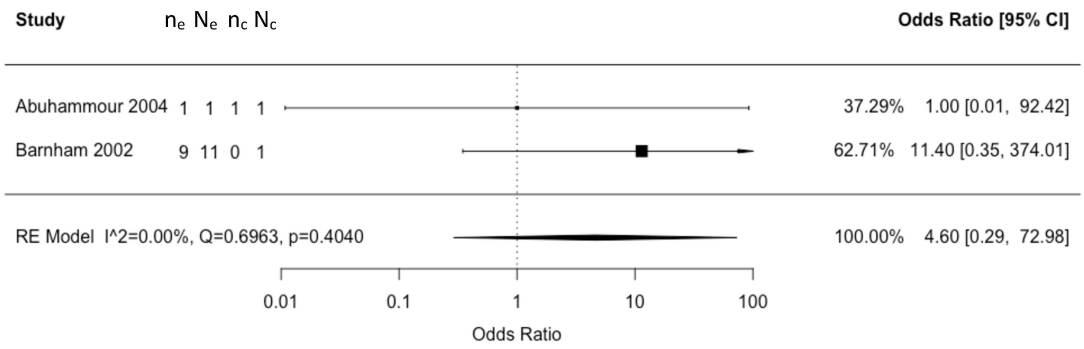
This note pertains to meta-analyses of the outcome ICU admission that include the study Stegmayr 1992. For this study, 10 of 11 STSS patients were admitted to the ICU. Despite no explicit mention of which one patient was not admitted to the ICU, our interpretation of the clinical profiles and patient-level data allowed us to deduce that patient 1 likely did not require ICU admission. We reached out to the corresponding author to confirm if our interpretation was correct, but did not receive a response.

1. Sex: male vs female (reference)

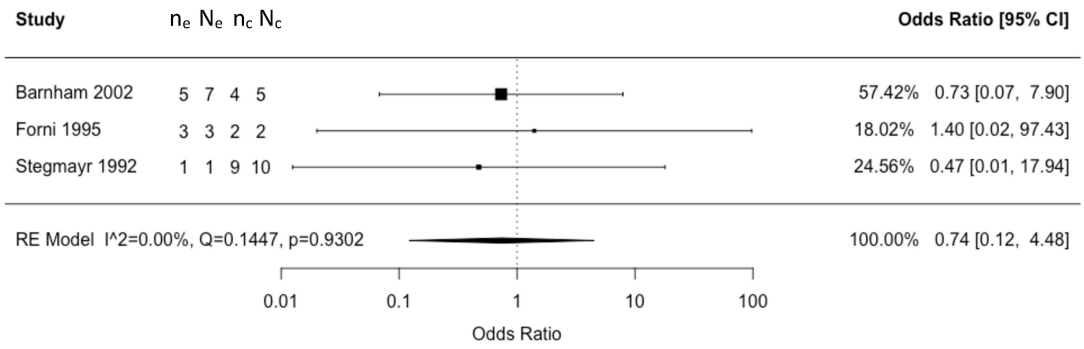




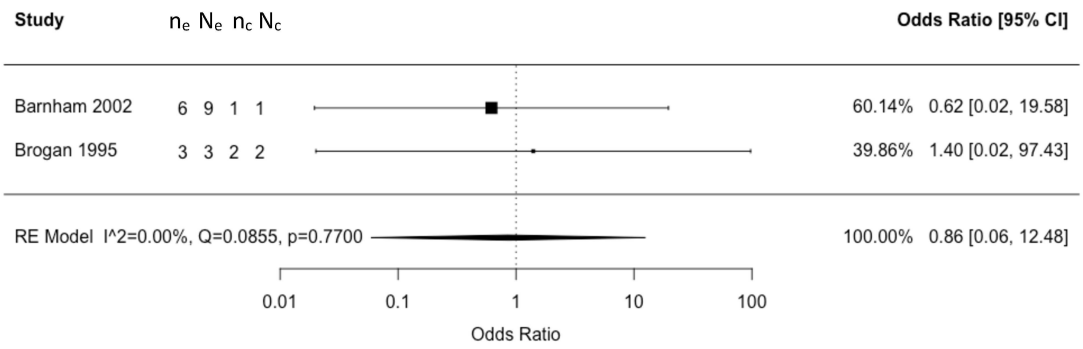
2. Any antibiotic: yes vs no (reference)

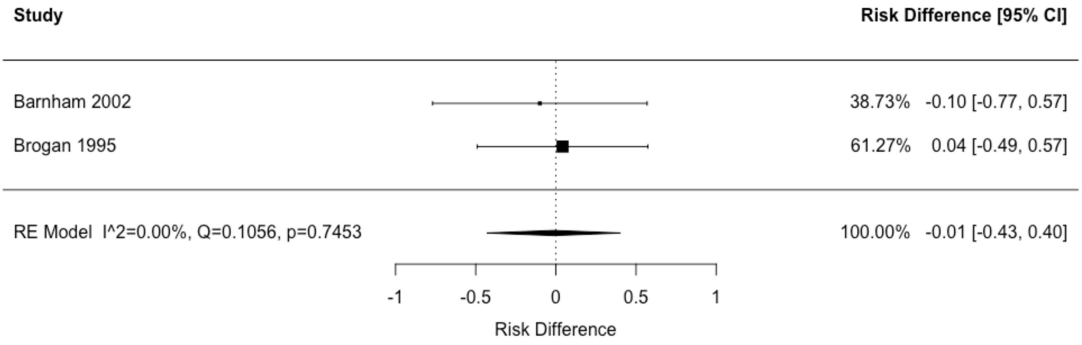


3. Necrotizing fasciitis: yes vs no (reference)

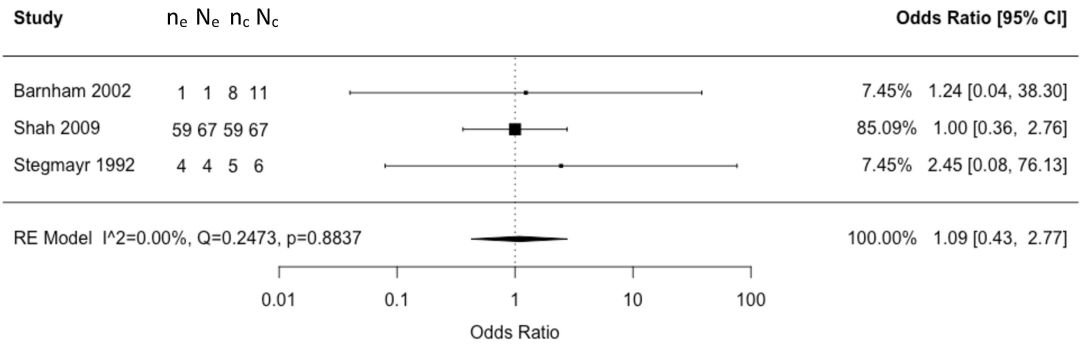


4. NSAIDs: yes vs no (reference)

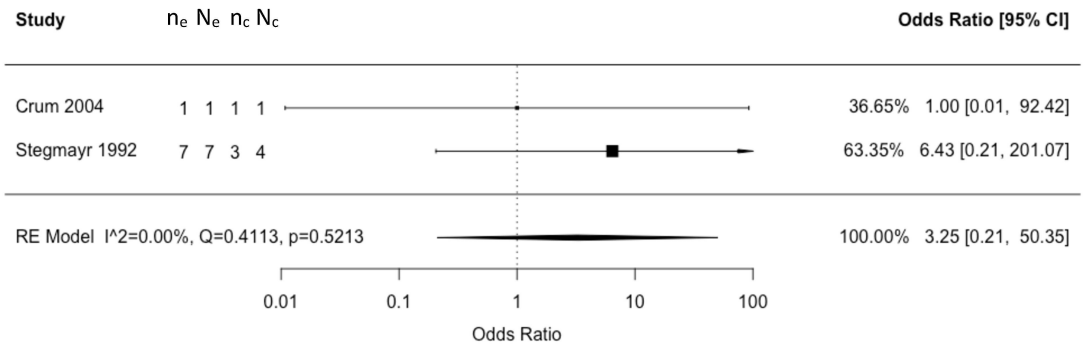




5. IVIG in all STSS patients: yes vs no (reference)

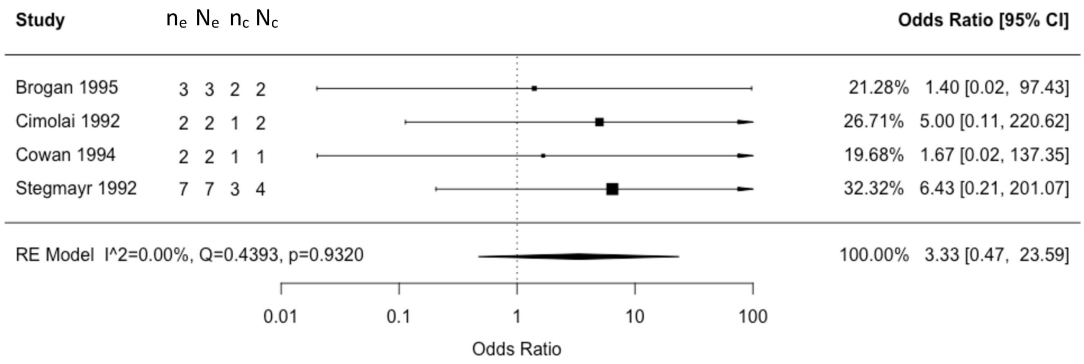


6. Hemodialysis: yes vs no (reference)

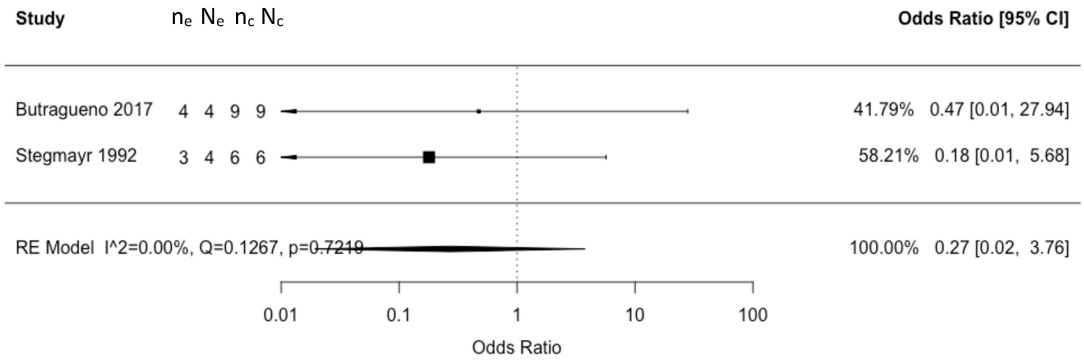


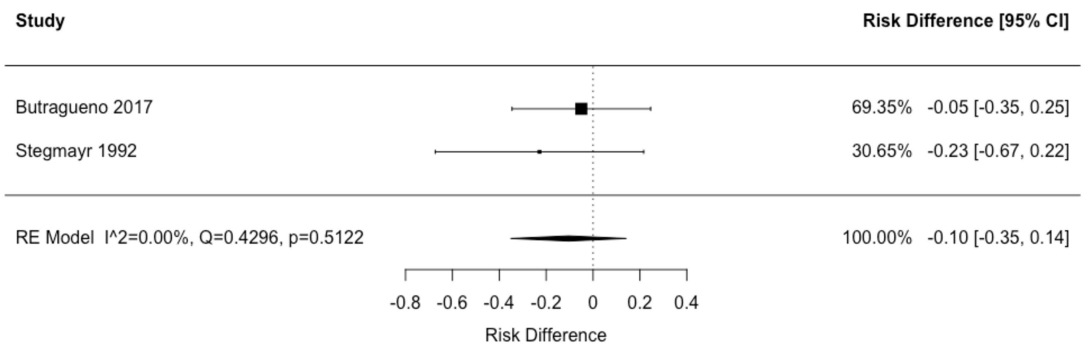
Clinical cure or improvement

1. Sex: male vs female (reference)

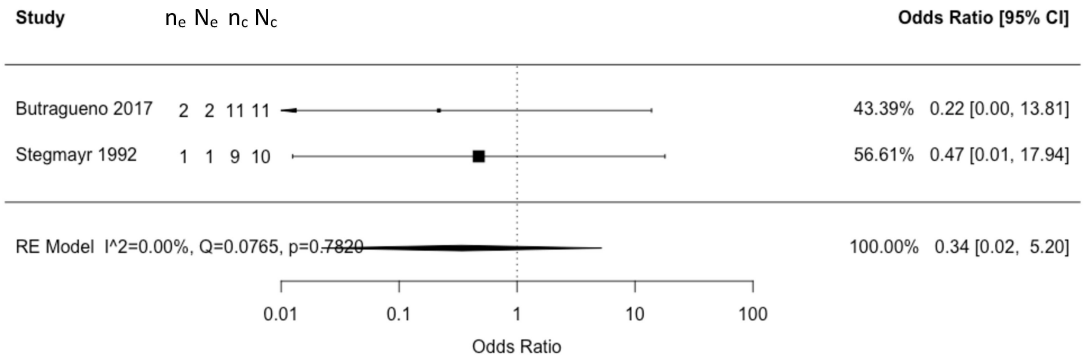


2. IVIG in all STSS patients: yes vs no (reference)

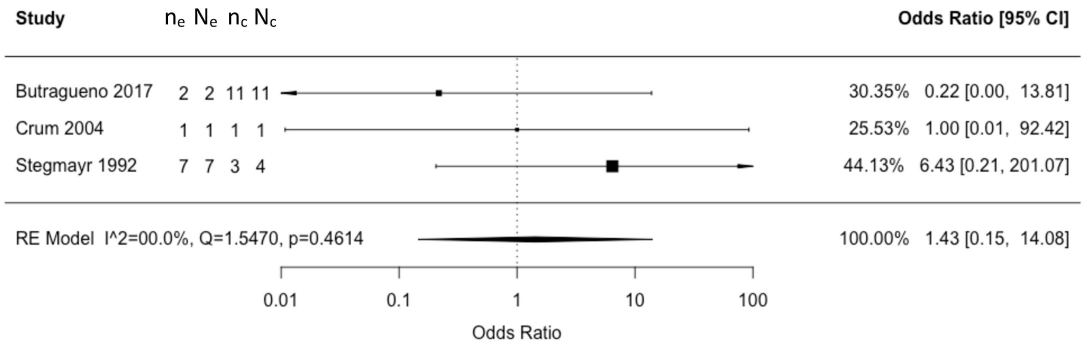


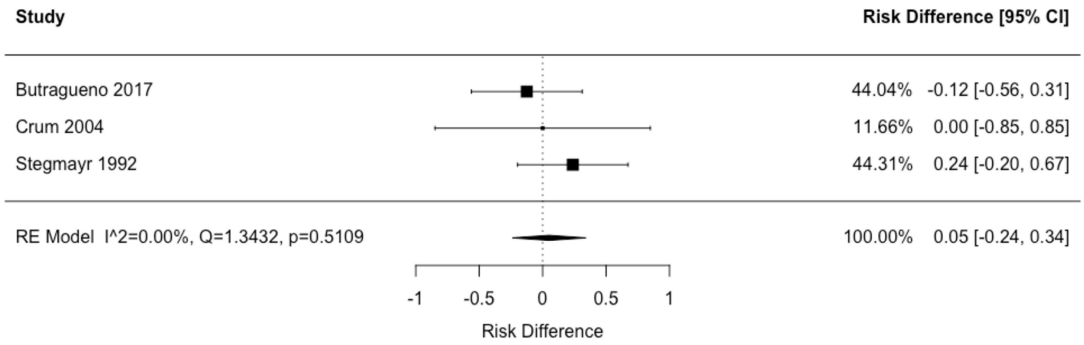


3. Necrotizing fasciitis: yes vs no (reference)



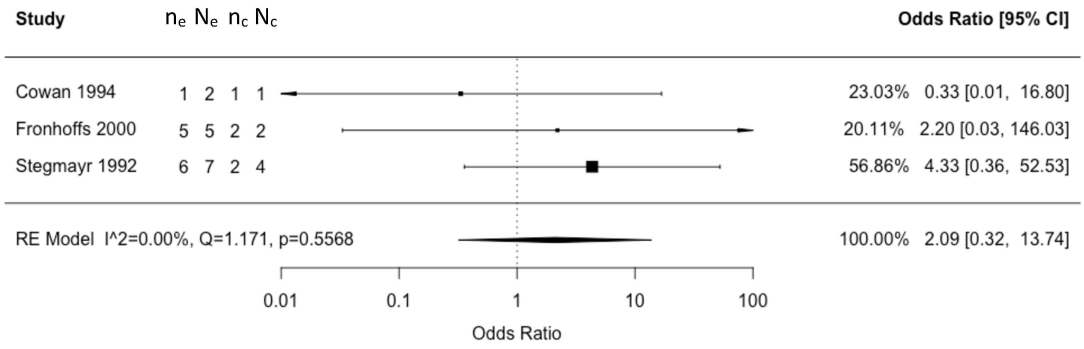
4. Hemodialysis: yes vs no (reference)

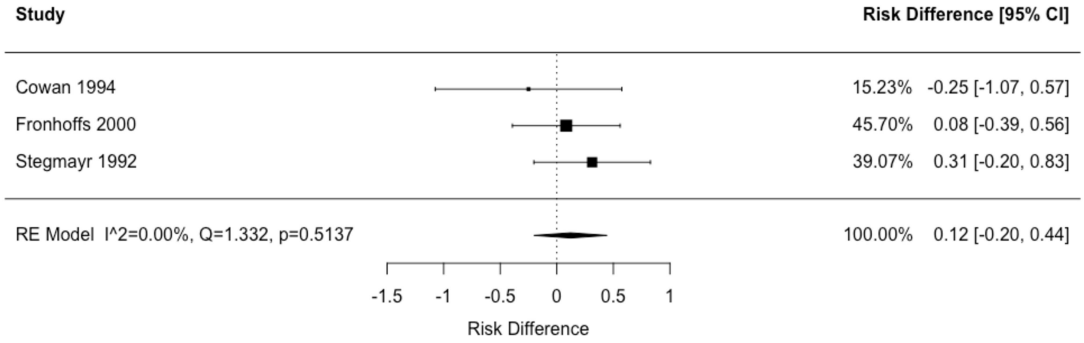




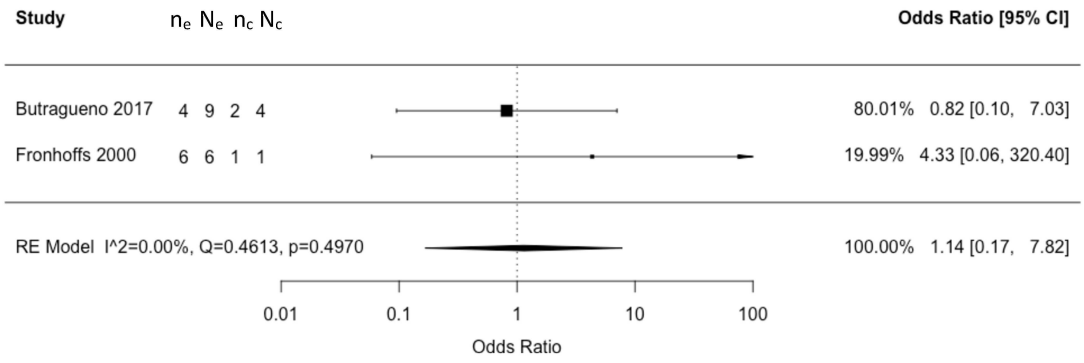
Mechanical ventilation

1. Sex: male vs female (reference)

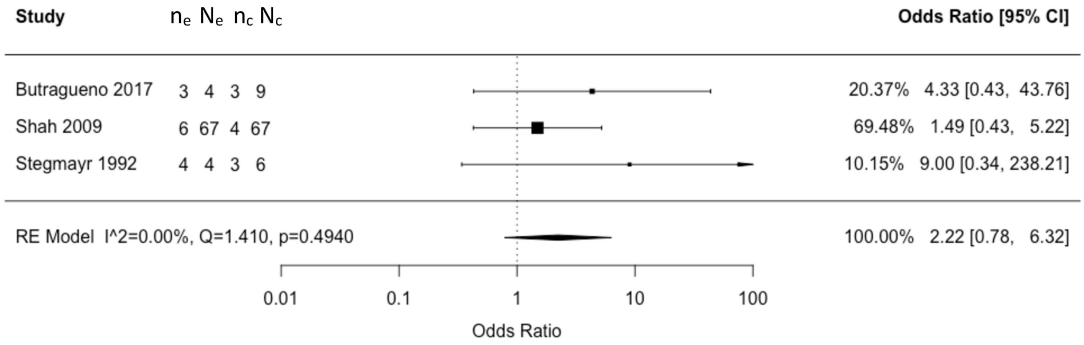




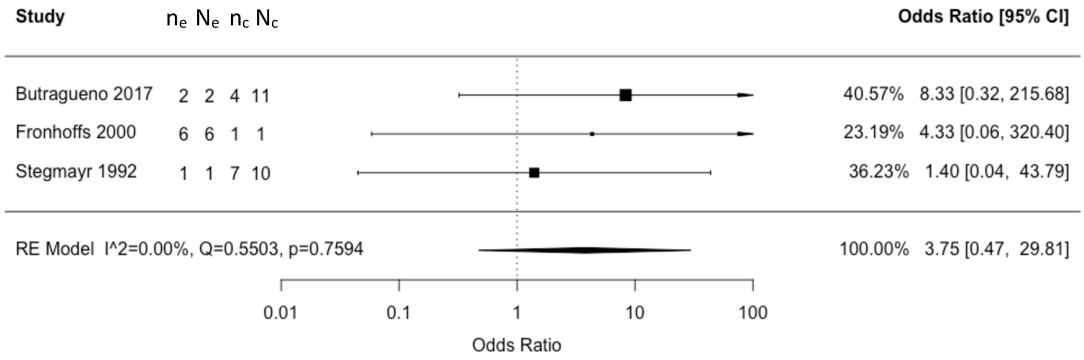
2. Acute renal failure: yes vs no (reference)



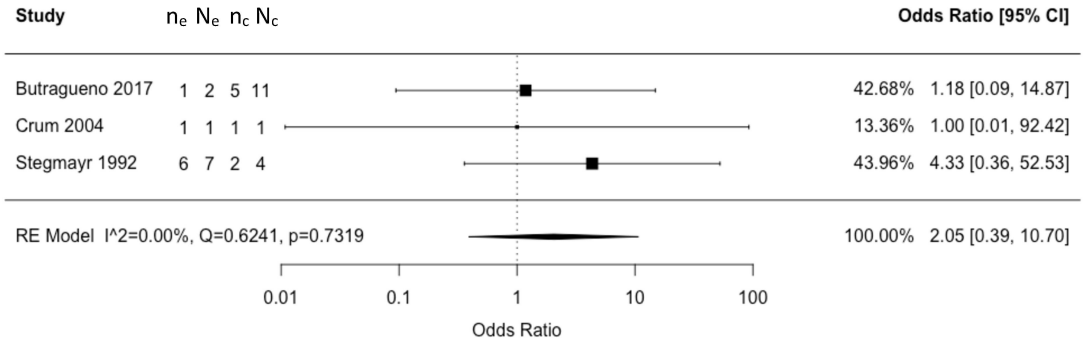
3. IVIG in all STSS patients: yes vs no (reference)



4. Necrotizing fasciitis: yes vs no (reference)

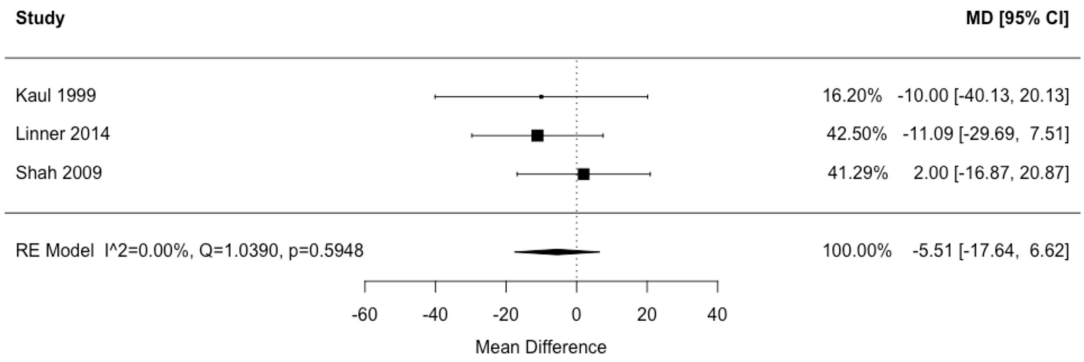


5. Hemodialysis: yes vs no (reference)



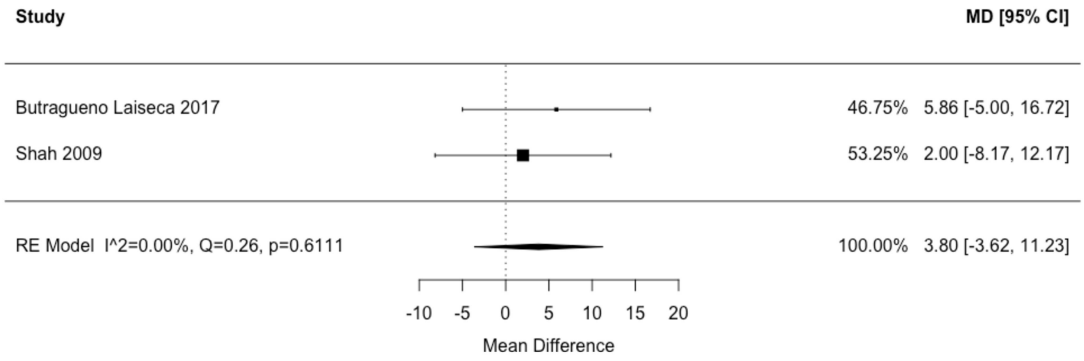
Hospital length-of-stay

1. IVIG: yes vs no (reference)



ICU length-of-stay

1. IVIG: yes vs no (reference)



Description of studies ineligible for meta-analysis by outcome

Mortality

Prognostic factor of interest	N reporting	N analyzed	Reasons for exclusion from meta-analysis
acute renal failure	4	4	
age	28	5	n=17 case-series with <10 patients, precluding the aggregation of patient-level data; n=6 study population consisted of patients all within same age category
antibiotic	3	3	
clindamycin	4	4	
early hypotension	0	0	
emmtype	7	0	n=7 variability in reporting of molecular characteristics and comparators
hemodialysis	4	4	
immunocompromised	5	4	n=1 insufficient data for meta-analysis (only p value reported)
IVIG in all STSS patients	9	9	
IVIG in clindamycin-treated patients	6	6	
NF	10	10	
NSAIDs	4	4	
sex	12	12	
timetoantibiotic	1	0	Meta-analysis precluded with only one study

(P)ICU admission

Prognostic factor of interest	N reporting	N analyzed	Reasons for exclusion from meta-analysis
acute renal failure	1	0	Meta-analysis precluded with only one study
age	9	0	n=5 case-series with <10 patients, precluding the aggregation of patient-level data; n=3 study population consisted of patients all within same age category; n=1 eligible for analysis, but meta-analysis precluded with only one study
antibiotic	2	2	
clindamycin	1	0	Meta-analysis precluded with only one study
early hypotension	0	0	
emmtype	2	0	n=2 variability in reporting of molecular characteristics and comparators
hemodialysis	2	2	
immunocompromised	1	0	Meta-analysis precluded with only one study
IVIG in all STSS patients	3	3	
IVIG in clindamycin-treated patients	0	0	
NF	3	3	
NSAIDs	2	2	
sex	3	3	
timetoantibiotic	0	0	

Clinical cure or improvement

Prognostic factor of interest	N reporting	N analyzed	Reasons for exclusion from meta-analysis
acute renal failure	1	0	Meta-analysis precluded with only one study
age	8	0	n=6 case-series with <10 patients, precluding the aggregation of patient-level data; n=2 study population consisted of patients all within same age category
antibiotic	1	0	Meta-analysis precluded with only one study
clindamycin	1	0	Meta-analysis precluded with only one study
earlyhypotension	0	0	
emmtype	0	0	
hemodialysis	3	3	
immunocompromised	0	0	
IVIG in all STSS patients	2	2	
IVIG in clindamycin-treated patients	0	0	
NF	2	2	
NSAIDs	1	0	Meta-analysis precluded with only one study
sex	4	4	
timetoantibiotic	0	0	

Mechanical ventilation

Prognostic factor of interest	N reporting	N analyzed	Reasons for exclusion from meta-analysis
acute renal failure	2	2	
age	5	0	n=3 case-series with <10 patients, precluding the aggregation of patient-level data; n=2 study population consisted of patients all within same age category
antibiotic	0	0	
clindamycin	1	0	Meta-analysis precluded with only one study
earlyhypotension	0	0	
emmtype	0	0	
hemodialysis	3	3	
immunocompromised	1	0	Meta-analysis precluded with only one study
IVIG in all STSS patients	3	3	
IVIG in clindamycin-treated patients	0	0	
NF	3	3	
NSAIDs	1	0	Meta-analysis precluded with only one study
sex	3	3	
timetoantibiotic	0	0	

Hospital length-of-stay

Prognostic factor of interest	N reporting	N analyzed	Reasons for exclusion from meta-analysis
acute renal failure	0	0	
age	2	0	n=2 case-series with <10 patients, precluding the aggregation of patient-level data
antibiotic	0	0	
clindamycin	0	0	
early hypotension	0	0	
emmtype	0	0	
hemodialysis	0	0	
immunocompromised	0	0	
IVIG in all STSS patients	3	3	
IVIG in clindamycin-treated patients	0	0	
NF	0	0	
NSAIDs	1	0	Meta-analysis precluded with only one study
sex	1	0	Meta-analysis precluded with only one study
timetoantibiotic	0	0	

Duration of mechanical ventilation

Prognostic factor of interest	N reporting	N analyzed	Reasons for exclusion from meta-analysis
acute renal failure	0	0	
age	0	0	
antibiotic	0	0	
clindamycin	0	0	
early hypotension	0	0	
emmtype	0	0	
hemodialysis	0	0	
immunocompromised	0	0	
IVIG in all STSS patients	1	0	Meta-analysis precluded with only one study
IVIG in clindamycin-treated patients	0	0	
NF	0	0	
NSAIDs	0	0	
sex	0	0	
timetoantibiotic	0	0	

ICU length-of-stay

Prognostic factor of interest	N reporting	N analyzed	Reasons for exclusion from meta-analysis
acute renal failure	0	0	
age	3	0	n=2 case-series with <10 patients, precluding the aggregation of patient-level data; n=1 study population consisted of patients all within same age category
antibiotic	0	0	
clindamycin	1	0	Meta-analysis precluded with only one study
early hypotension	0	0	
emmtype	0	0	
hemodialysis	0	0	
immunocompromised	0	0	
IVIG in all STSS patients	2	2	
IVIG in clindamycin-treated patients	0	0	
NF	1	0	Meta-analysis precluded with only one study
NSAIDs	1	0	Meta-analysis precluded with only one study
sex	2	0	n=1 study with one person in each group > cannot calculate mean; n=1 meta-analysis precluded with only one study
timetoantibiotic	0	0	

Change in SOFA score from baseline

Prognostic factor of interest	N reporting	N analyzed	Reasons for exclusion from meta-analysis
acute renal failure	0	0	
age	0	0	
antibiotic	0	0	
clindamycin	0	0	
early hypotension	0	0	
emmtype	0	0	
hemodialysis	0	0	
immunocompromised	0	0	
IVIG in all STSS patients	1	0	Meta-analysis precluded with only one study
IVIG in clindamycin-treated patients	0	0	
NF	0	0	
NSAIDs	0	0	
sex	0	0	
timetoantibiotic	0	0	

Functional status

Prognostic factor of interest	N reporting	N analyzed	Reasons for exclusion from meta-analysis
acute renal failure	0	0	
age	0	0	
antibiotic	0	0	
clindamycin	0	0	
early hypotension	0	0	
emmtype	0	0	
hemodialysis	0	0	
immunocompromised	0	0	
IVIG in all STSS patients	0	0	
IVIG in clindamycin-treated patients	0	0	
NF	0	0	
NSAIDs	0	0	
sex	0	0	
timetoantibiotic	0	0	

Health related quality of life

Prognostic factor of interest	N reporting	N analyzed	Reasons for exclusion from meta-analysis
acute renal failure	0	0	
age	0	0	
antibiotic	0	0	
clindamycin	0	0	
early hypotension	0	0	
emmtype	0	0	
hemodialysis	0	0	
immunocompromised	0	0	
IVIG in all STSS patients	0	0	
IVIG in clindamycin-treated patients	0	0	
NF	0	0	
NSAIDs	0	0	
sex	0	0	
timetoantibiotic	0	0	

Cost

Prognostic factor of interest	N reporting	N analyzed	Reasons for exclusion from meta-analysis
acute renal failure	0	0	
age	0	0	
antibiotic	0	0	
clindamycin	0	0	
early hypotension	0	0	
emmtype	0	0	
hemodialysis	0	0	
immunocompromised	0	0	
IVIG in all STSS patients	1	0	Meta-analysis precluded with only one study
IVIG in clindamycin-treated patients	0	0	
NF	0	0	
NSAIDs	0	0	
sex	0	0	
timetoantibiotic	0	0	

Time to mortality

Prognostic factor of interest	N reporting	N analyzed	Reasons for exclusion from meta-analysis
acute renal failure	0	0	
age	0	0	
antibiotic	0	0	
clindamycin	0	0	
early hypotension	0	0	
emmtype	0	0	
hemodialysis	0	0	
immunocompromised	0	0	
IVIG in all STSS patients	2	0	n=1 study with one person in each group > cannot calculate mean; n=1 meta-analysis precluded with only one study
IVIG in clindamycin-treated patients	2	0	n=1 study with one person in each group > cannot calculate mean; n=1 meta-analysis precluded with only one study
NF	0	0	
NSAIDs	0	0	
sex	0	0	
timetoantibiotic	0	0	

Time to clinical improvement or resolution of shock

Prognostic factor of interest	N reporting	N analyzed	Reasons for exclusion from meta-analysis
acute renal failure	0	0	
age	0	0	
antibiotic	0	0	
clindamycin	0	0	
early hypotension	0	0	
emmtype	0	0	
hemodialysis	0	0	
immunocompromised	0	0	
IVIG in all STSS patients	1	0	Meta-analysis precluded with only one study
IVIG in clindamycin-treated patients	1	0	Meta-analysis precluded with only one study
NF	0	0	
NSAIDs	0	0	
sex	0	0	
timetoantibiotic	0	0	